

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 93 on 01/08/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTA4MDExMTI5MjlfNzAyMw%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

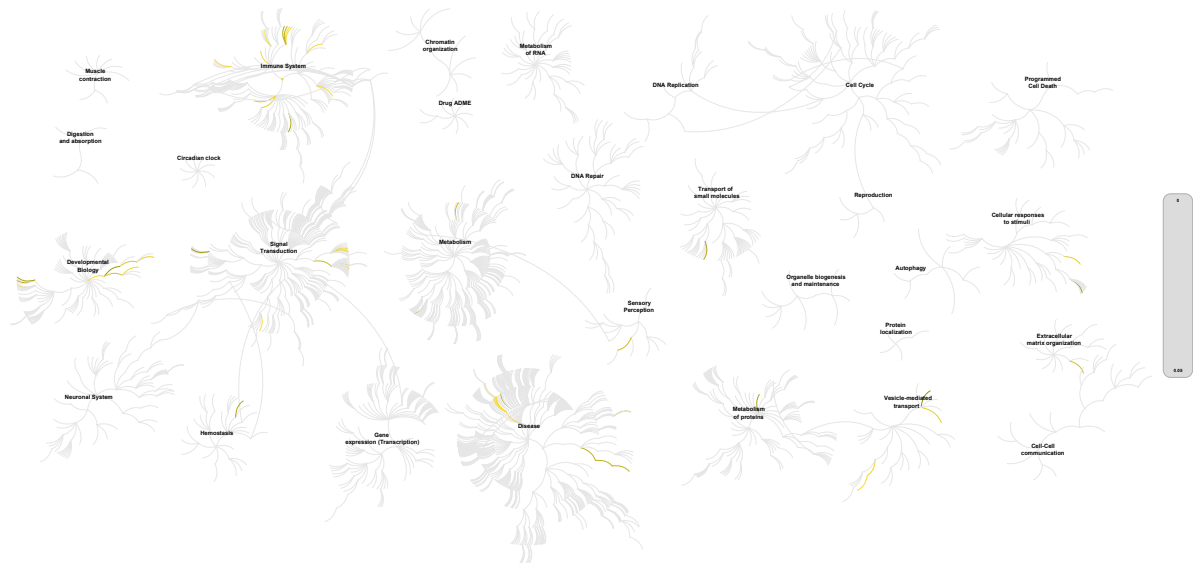
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamani-Hochberg method. [↗](#)
- 26 out of 28 identifiers in the sample were found in Reactome, where 756 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTA4MDExMTI5MjlfNzAyMw%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

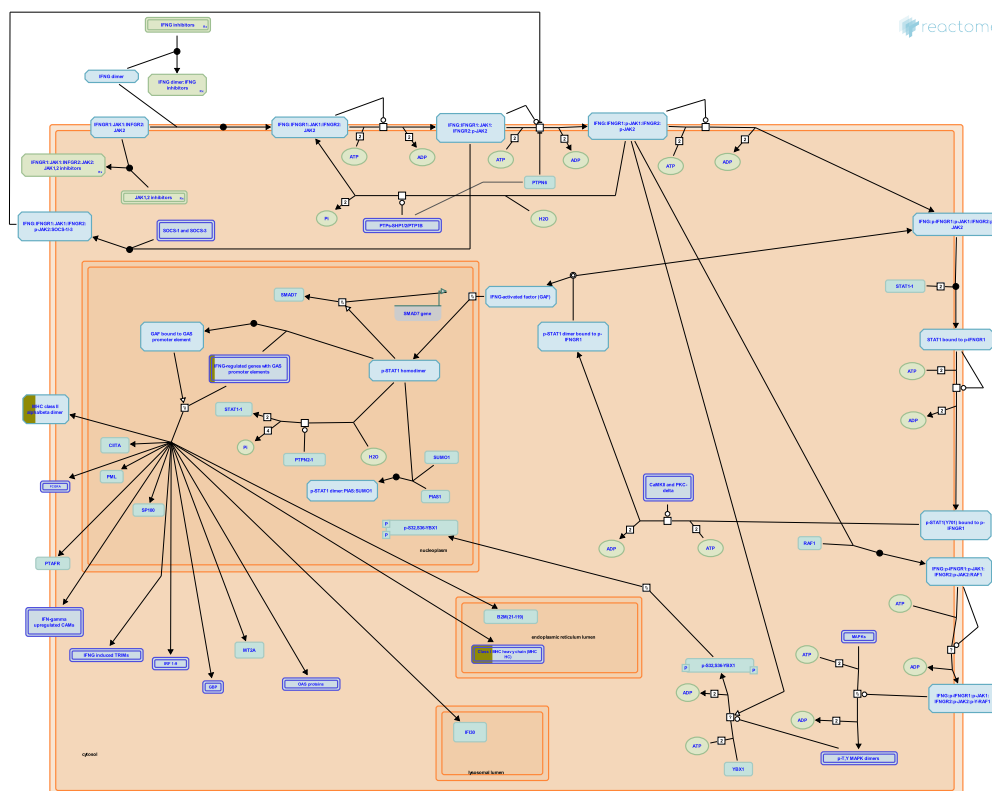
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Interferon gamma signaling	11 / 625	0.025	7.20e-06	0.006	2 / 23	0.001
Translocation of ZAP-70 to Immunological synapse	3 / 40	0.002	1.89e-04	0.061	4 / 4	2.57e-04
Signaling by high-kinase activity BRAF mutants	3 / 44	0.002	2.49e-04	0.061	4 / 6	3.86e-04
Signaling by RAF1 mutants	3 / 49	0.002	3.41e-04	0.061	4 / 7	4.50e-04
MAP2K and MAPK activation	3 / 51	0.002	3.83e-04	0.061	8 / 12	7.72e-04
Co-inhibition by PD-1	3 / 54	0.002	4.52e-04	0.061	1 / 5	3.21e-04
Endosomal/Vacuolar pathway	2 / 15	6.08e-04	7.96e-04	0.067	3 / 4	2.57e-04
Phosphorylation of CD3 and TCR zeta chains	3 / 69	0.003	9.16e-04	0.067	5 / 7	4.50e-04
Signaling by BRAF and RAF1 fusions	3 / 73	0.003	0.001	0.067	4 / 5	3.21e-04
Signaling downstream of RAS mutants	3 / 73	0.003	0.001	0.067	4 / 7	4.50e-04
Signaling by moderate kinase activity BRAF mutants	3 / 73	0.003	0.001	0.067	4 / 7	4.50e-04
Paradoxical activation of RAF signaling by kinase inactive BRAF	3 / 73	0.003	0.001	0.067	4 / 7	4.50e-04
Signaling by RAS mutants	3 / 73	0.003	0.001	0.067	4 / 9	5.79e-04
Developmental Cell Lineages	5 / 292	0.012	0.001	0.068	4 / 17	0.001
Developmental Lineage of Pancreatic Ductal Cells	3 / 77	0.003	0.001	0.068	2 / 2	1.29e-04
Formation of annular gap junctions	2 / 21	8.51e-04	0.002	0.077	2 / 2	1.29e-04
Interleukin-4 and Interleukin-13 signaling	6 / 473	0.019	0.002	0.086	3 / 47	0.003
RHO GTPases Activate WASPs and WAVES	3 / 97	0.004	0.002	0.109	4 / 10	6.43e-04
MHC class II antigen presentation	5 / 361	0.015	0.003	0.128	25 / 26	0.002
Oncogenic MAPK signaling	3 / 112	0.005	0.004	0.142	24 / 46	0.003
Interaction between L1 and Ankyrins	2 / 33	0.001	0.004	0.142	2 / 4	2.57e-04
Gap junction degradation	2 / 40	0.002	0.005	0.195	4 / 4	2.57e-04
Immune System	42 / 8,019	0.325	0.007	0.231	156 / 1,733	0.111
RHOF GTPase cycle	2 / 46	0.002	0.007	0.231	1 / 3	1.93e-04
Developmental Lineage of Mammary Gland Myoepithelial Cells	2 / 47	0.002	0.007	0.231	2 / 2	1.29e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Interferon gamma signaling (R-HSA-877300)



Interferon-gamma (IFN-gamma) belongs to the type II interferon family and is secreted by activated immune cells-primarily T and NK cells, but also B-cells and APC. IFNG exerts its effect on cells by interacting with the specific IFN-gamma receptor (IFNGR). IFNGR consists of two chains, namely IFNGR1 (also known as the IFNGR alpha chain) and IFNGR2 (also known as the IFNGR beta chain). IFNGR1 is the ligand binding receptor and is required but not sufficient for signal transduction, whereas IFNGR2 do not bind IFNG independently but mainly plays a role in IFNG signaling and is generally the limiting factor in IFNG responsiveness. Both IFNGR chains lack intrinsic kinase/phosphatase activity and thus rely on other signaling proteins like Janus-activated kinase 1 (JAK1), JAK2 and Signal transducer and activator of transcription 1 (STAT-1) for signal transduction. IFNGR complex in its resting state is a preformed tetramer and upon IFNG association undergoes a conformational change. This conformational change induces the phosphorylation and activation of JAK1, JAK2, and STAT1 which in turn induces genes containing the gamma-interferon activation sequence (GAS) in the promoter.

References

Izotova LS, Garotta G, Muthukumaran G, Kotenko SV, Cook JR & Pestka S (1997). The interferon gamma (IFN-gamma) receptor: a paradigm for the multichain cytokine receptor. Cytokine Growth Factor Rev, 8, 189-206. [🔗](#)

Gough DJ, Levy DE, Clarke CJ & Johnstone RW (2008). IFNgamma signaling-does it mean JAK-STAT?. Cytokine Growth Factor Rev, 19, 383-94. [↗](#)

Aguet M, Bach EA & Schreiber RD (1997). The IFN gamma receptor: a paradigm for cytokine receptor signaling. Annu Rev Immunol, 15, 563-91. [↗](#)

Schroder K, Ravasi T, Hume DA & Hertzog PJ (2004). Interferon-gamma: an overview of signals, mechanisms and functions. J Leukoc Biol, 75, 163-89. [↗](#)

Edit history

Date	Action	Author
2010-06-08	Edited	Garapati P V
2010-06-08	Authored	Garapati P V
2010-06-11	Created	Garapati P V
2010-08-17	Reviewed	Abdul-Sater AA, Schindler C
2024-11-15	Reviewed	Matthews L

5 submitted entities found in this pathway, mapping to 10 Reactome entities

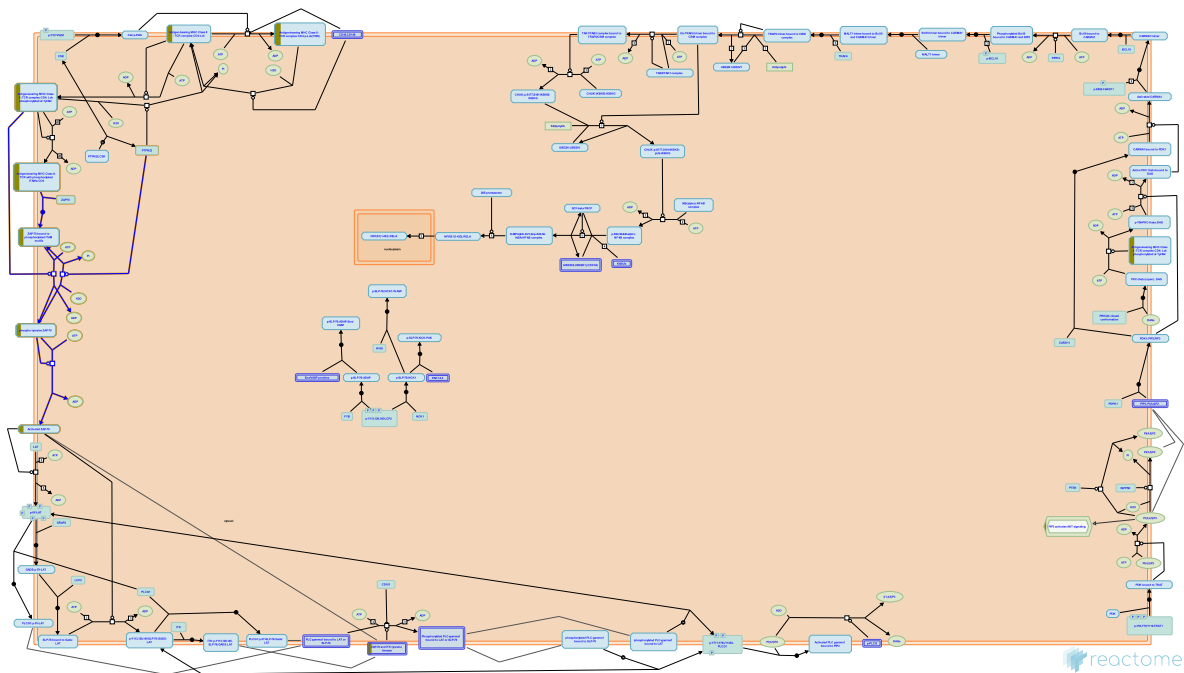
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HLA-DQB1	P01920	HLA-DRB5	Q30154		

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
HLA-A	ENSG00000206503	HLA-B	ENSG00000234745	HLA-DQA1	ENSG00000196735
HLA-DQB1	ENSG00000179344	HLA-DRB5	ENSG00000198502		

Interactors found in this pathway (3)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
APOE	P02649	P02795	HLA-A	P04439	P61769
HLA-B	P01889	P61769			

2. Translocation of ZAP-70 to Immunological synapse (R-HSA-202430)



Cellular compartments: plasma membrane.

The dual phosphorylated ITAMs recruit SYK kinase ZAP70 via their tandem SH2 domains (step 8). ZAP70 subsequently undergoes phosphorylation on multiple tyrosine residues for further activation. ZAP70 includes both positive and negative regulatory sites. Tyrosine 493 is a conserved regulatory site found within the activation loop of the kinase domain. This site has shown to be a positive regulatory site required for ZAP70 kinase activity and is phosphorylated by LCK (step 9). This phosphorylation contributes to the active conformation of the catalytic domain. Later ZAP70 undergoes trans-autophosphorylation at Y315 and Y319 (step 10). These sites appear to be positive regulatory sites. ZAP70 achieves its full activation after the trans-autophosphorylation. Activated ZAP70 along with LCK phosphorylates the multiple tyrosine residues in the adaptor protein LAT (step 11). PTPN22 can dephosphorylate and inhibit ZAP70 activity to downregulate TCR signaling (step 12).

References

van Leeuwen JE & Samelson LE (1999). T cell antigen-receptor signal transduction. Curr Opin Immunol, 11, 242-8. [↗](#)

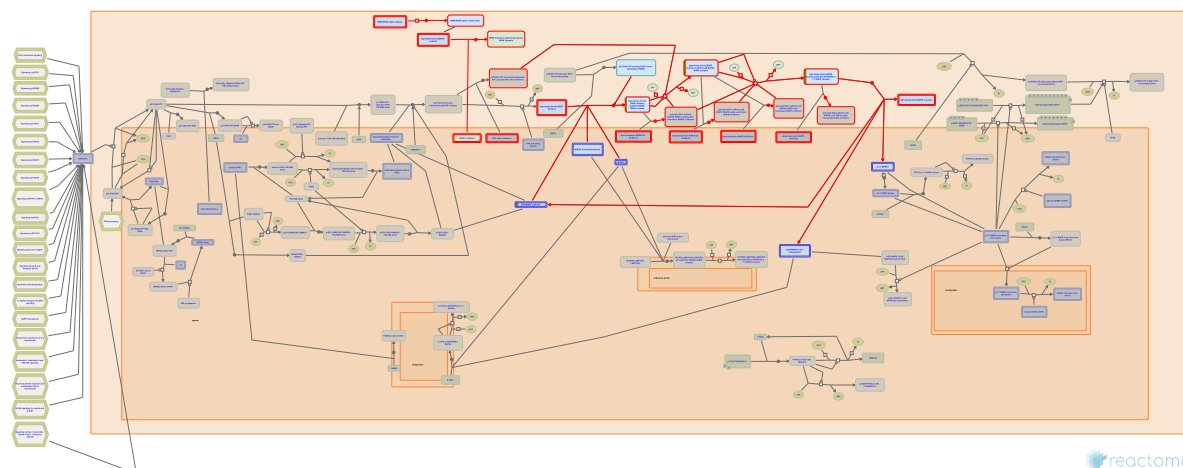
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Date	Action	Author
2007-10-29	Created	Garapati P V
2008-01-24	Authored	Rudd C.E., Garapati P V, de Bono B
2008-02-26	Reviewed	Trowsdale J
2016-05-10	Revised	Stanford S, Bottini N

3 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HLA-DQA1	P01909	HLA-DQB1	P01920	HLA-DRB5	Q30154

3. Signaling by high-kinase activity BRAF mutants (R-HSA-6802948)



Diseases: cancer.

BRAF is mutated in about 8% of human cancers, with high prevalence in hairy cell leukemia, melanoma, papillary thyroid and ovarian carcinomas, colorectal cancer and a variety of other tumors (Davies et al, 2002; reviewed in Samatar and Poulikakos, 2014). Most BRAF mutations fall in the activation loop region of the kinase or the adjacent glycine rich region. These mutations promote increased kinase activity either by mimicking the effects of activation loop phosphorylations or by promoting the active conformation of the enzyme (Davies et al, 2002; Wan et al, 2004). Roughly 90% of BRAF mutants are represented by the single missense mutation BRAF V600E (Davies et al, 2002; Wan et al, 2004). Other highly active kinase mutants of BRAF include BRAF G469A and BRAF T599dup. G469 is in the glycine rich region of the kinase domain which plays a role in orienting ATP for catalysis, while T599 is one of the two conserved regulatory phosphorylation sites of the activation loop. Each of these mutants has highly enhanced basal kinase activities, phosphorylates MEK and ERK in vitro and in vivo and is transforming when expressed in vivo (Davies et al, 2002; Wan et al, 2004; Eisenhardt et al, 2011). Further functional characterization shows that these highly active mutants are largely resistant to disruption of the BRAF dimer interface, suggesting that they are able to act as monomers (Roring et al, 2012; Brummer et al, 2006; Freeman et al, 2013; Garnett et al, 2005). Activating BRAF mutations occur for the most part independently of RAS activating mutations, and RAS activity levels are generally low in BRAF mutant cells. Moreover, the kinase activity of these mutants is only slightly elevated by coexpression of G12V KRAS, and biological activity of the highly active BRAF mutants is independent of RAS binding (Brummer et al, 2006; Wan et al, 2004; Davies et al, 2002; Garnett et al, 2005). Although BRAF V600E is inhibited by RAF inhibitors such as vemurafenib, resistance frequently develops, in some cases mediated by the expression of a splice variant that lacks the RAS binding domain and shows elevated dimerization compared to the full length V600E mutant (Poulikakos et al, 2011; reviewed in Lito et al, 2013).

References

- Poulikakos PI & Samatar AA (2014). Targeting RAS-ERK signalling in cancer: promises and challenges. *Nat Rev Drug Discov*, 13, 928-42. [🔗](#)
- Niculescu-Duvaz D, Lee S, Marais R, Marshall CJ, Barford D, Jones CM, ... Roe SM (2004). Mechanism of activation of the RAF-ERK signaling pathway by oncogenic mutations of B-RAF. *Cell*, 116, 855-67. [🔗](#)

Graeber TG, Dahlman KB, Persaud Y, Sosman JA, Kong X, Ribas A, ... Salton M (2011). RAF inhibitor resistance is mediated by dimerization of aberrantly spliced BRAF(V600E). Nature, 480, 387-90.



Lito P, Solit DB & Rosen N (2013). Tumor adaptation and resistance to RAF inhibitors. Nat. Med., 19, 1401-9.



Olbrich H, Röring M, Janzarik W, Witt H, Anh TN, Korshunov A, ... Eisenhardt AE (2011). Functional characterization of a BRAF insertion mutant associated with pilocytic astrocytoma. Int. J. Cancer, 129, 2297-303.



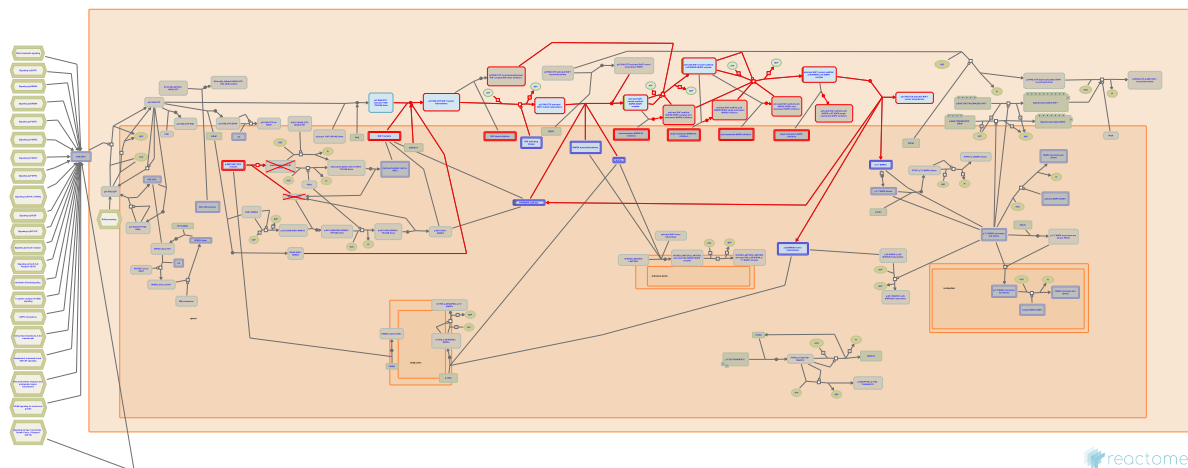
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Date	Action	Author
2015-08-10	Edited	Rothfels K
2015-08-10	Authored	Rothfels K
2015-10-02	Created	Rothfels K
2016-08-05	Reviewed	Stephens RM

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

4. Signaling by RAF1 mutants (R-HSA-9656223)



Diseases: Costello syndrome, LEOPARD syndrome, cancer, Noonan syndrome, hypertrophic cardiomyopathy.

RAF1, also known as CRAF, is mutated in a number of germline RASopathies including Noonan Syndrome, Costello Syndrome and others, and also at low frequency in a number of cancers (reviewed in Rauen, 2013; Samatar and Poulikakos, 2015). Activating mutations cluster around conserved region 2 (CR2) which is required for regulation of the protein and the activation segment in CR3 (reviewed in Rauen, 2013).

References

Poulikakos PI & Samatar AA (2014). Targeting RAS-ERK signalling in cancer: promises and challenges. Nat Rev Drug Discov, 13, 928-42. [🔗](#)

Rauen KA (2013). The RASopathies. Annu Rev Genomics Hum Genet, 14, 355-69. [🔗](#)

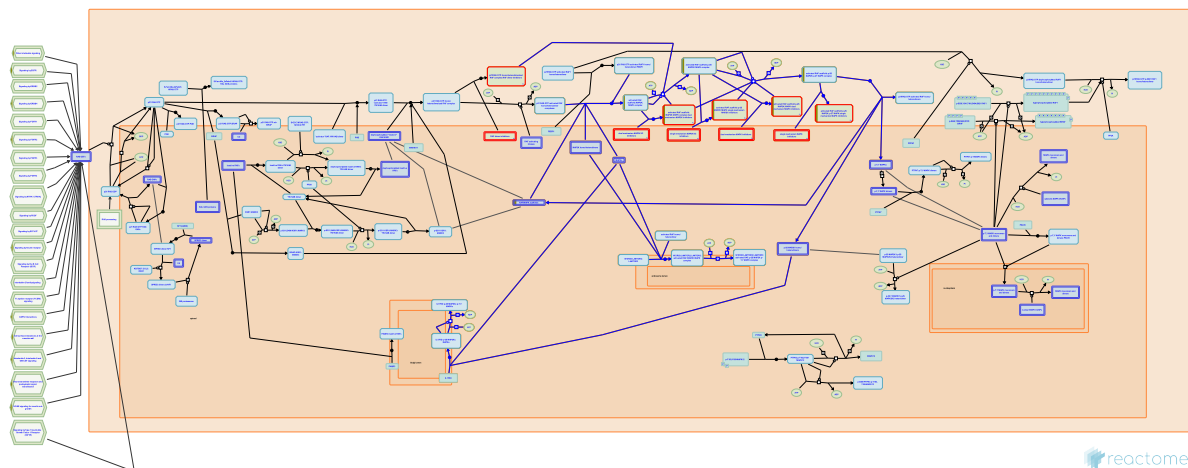
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Date	Action	Author
2019-07-27	Created	Rothfels K
2019-10-25	Authored	Rothfels K
2020-05-04	Reviewed	Gavathiotis E
2020-05-26	Edited	Rothfels K

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

5. MAP2K and MAPK activation (R-HSA-5674135)



Activated RAF proteins are restricted substrate kinases whose primary downstream targets are the two MAP2K proteins, MAPK2K1 and MAP2K2 (also known as MEK1 and MEK2) (reviewed in Roskoski, 2010, Roskoski, 2012a). Phosphorylation of the MAPK2K activation loop primes them to phosphorylate the primary effector of the activated MAPK pathway, the two MAPK proteins MAPK3 and MAPK1 (also known as ERK1 and 2). Unlike their upstream counterparts, MAPK3 and MAPK1 catalyze the phosphorylation of hundreds of cytoplasmic and nuclear targets including transcription factors and regulatory molecules (reviewed in Roskoski, 2012b). Activation of MAP2K and MAPK proteins downstream of activated RAF generally occurs in the context of a higher order scaffolding complex that regulates the specificity and localization of the pathway (reviewed in Brown and Sacks, 2009; Matallanas et al, 2011).

References

Romano D, Matallanas D, Rauch J, Zebisch A, Birtwistle M, Kolch W & von Kriegsheim A (2011). Raf family kinases: old dogs have learned new tricks. *Genes Cancer*, 2, 232-60. [🔗](#)

Roskoski R Jr (2012). ERK1/2 MAP kinases: structure, function, and regulation. *Pharmacol. Res.*, 66, 105-43. [🔗](#)

Roskoski R Jr (2010). RAF protein-serine/threonine kinases: structure and regulation. *Biochem. Biophys. Res. Commun.*, 399, 313-7. [🔗](#)

Brown MD & Sacks DB (2009). Protein scaffolds in MAP kinase signalling. *Cell. Signal.*, 21, 462-9. [🔗](#)

Roskoski R Jr (2012). MEK1/2 dual-specificity protein kinases: structure and regulation. *Biochem. Biophys. Res. Commun.*, 417, 5-10. [🔗](#)

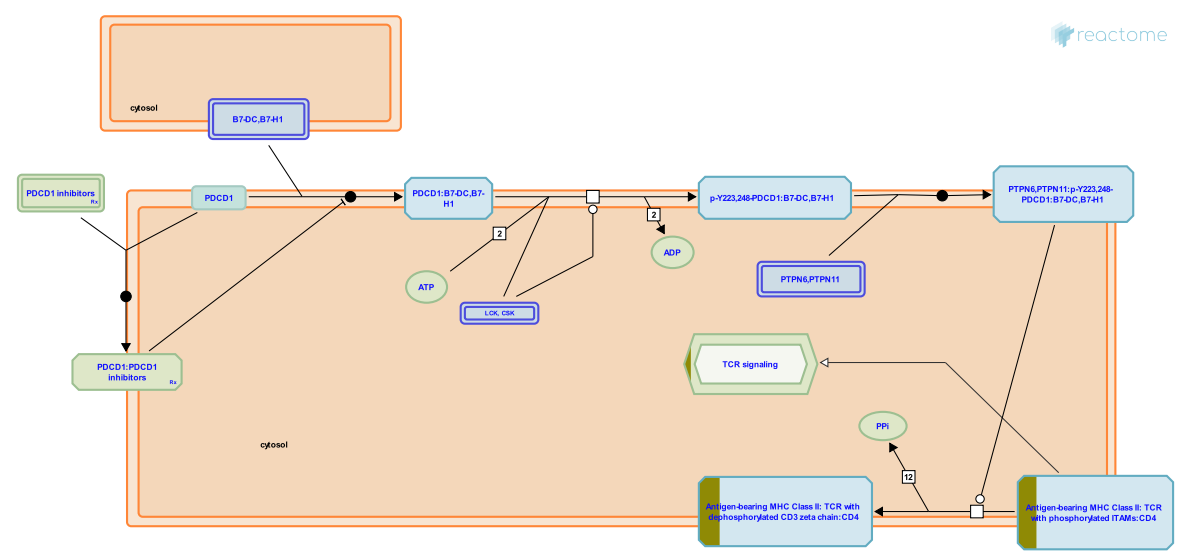
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Date	Action	Author
2015-02-10	Authored	Rothfels K
2015-02-11	Created	Rothfels K
2015-02-12	Edited	Rothfels K
2015-04-29	Reviewed	Roskoski R Jr

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

6. Co-inhibition by PD-1 (R-HSA-389948)



Cellular compartments: plasma membrane.

Programmed cell death protein 1 (PD-1) is a crucial inhibitory receptor that regulates T cell receptor (TCR) signaling and plays a vital role in maintaining immune homeostasis. PD-1 exerts its suppressive effects both directly, by inhibiting early activation events that are otherwise enhanced by co-stimulatory signals like CD28, and indirectly, by reducing interleukin-2 (IL-2) production, which is essential for T cell proliferation and survival. Upon ligation, PD-1 inhibits the expression of key survival and differentiation factors, such as Bcl-xL, and downregulates transcription factors that are central to effector T cell function, including GATA-3, T-bet, and Eomes. Mechanistically, PD-1 recruits the tyrosine phosphatases SHP-1 and SHP-2 to the immune synapse, leading to the dephosphorylation of critical signaling molecules like the CD3-zeta chain, PI3K, and AKT, thereby attenuating TCR signaling and inhibiting T cell activation and function (Keir et al. 2008).

References

Fife BT & Bluestone JA (2008). Control of peripheral T-cell tolerance and autoimmunity via the CTLA-4 and PD-1 pathways. *Immunol Rev*, 224, 166-82. [🔗](#)

Keir ME, Sharpe AH, Freeman GJ & Butte MJ (2008). PD-1 and its ligands in tolerance and immunity . *Annu Rev Immunol*, 26, 677-704. [🔗](#)

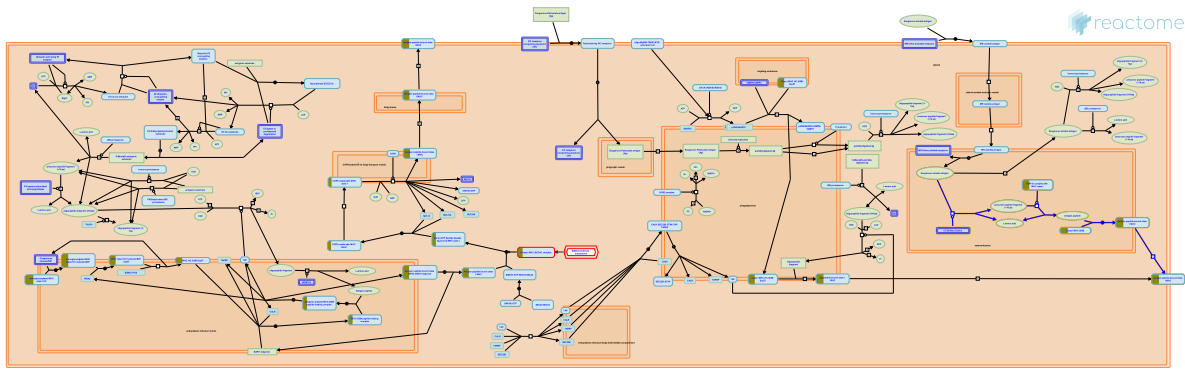
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2008-12-16	Edited	Garapati P V
2008-12-16	Authored	Garapati P V
2009-01-21	Created	Garapati P V
2009-06-01	Reviewed	Bluestone JA, Esensten J
2024-11-07	Edited	Tiwari K
2024-11-07	Reviewed	D'Eustachio P
2024-11-07	Authored	Garapati P V

3 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HLA-DQA1	P01909	HLA-DQB1	P01920	HLA-DRB5	Q30154

7. Endosomal/Vacuolar pathway (R-HSA-1236977)



Cellular compartments: early endosome.

Some antigens are cross-presented through a vacuolar mechanism that involves generation of antigenic peptides and their loading on to MHC-I molecules within the endosomal compartment in a proteasome and TAP-independent manner. Antigens within the endosome are processed by cathepsin S and other proteases into antigenic peptides. Loading of these peptides onto MHC-I molecules occurs directly within early and late endosomal compartments. Why certain antigens are cross-presented exclusively by the cytosolic pathway while others use the vacuolar pathway is unknown. It may be because some epitopes cannot be generated by endosomal proteolysis, or are completely destroyed. Alternatively, the physical form of the antigen may influence its accessibility to the endosomal or vacuolar pathways (Shen et al. 2004).

References

Banchereau J, Montes M, Mellman I, Xue Y, Connolly JE, Clayton S, ... Di Pucchio T (2008). Direct proteasome-independent cross-presentation of viral antigen by plasmacytoid dendritic cells on major histocompatibility complex class I. Nat Immunol, 9, 551-7. [🔗](#)

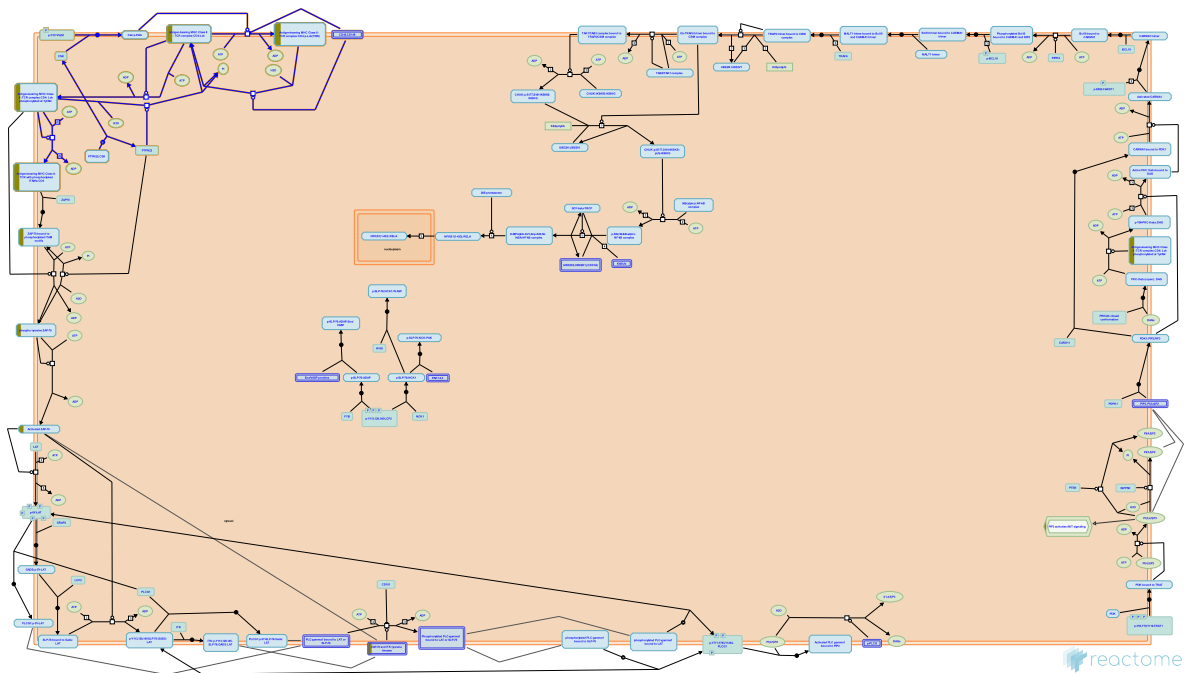
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2011-03-28	Authored	Garapati P V
2011-03-28	Created	Garapati P V
2011-05-13	Reviewed	Desjardins M, English L
2025-06-09	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
HLA-A	P04439	HLA-B	P01889

8. Phosphorylation of CD3 and TCR zeta chains (R-HSA-202427)



Cellular compartments: plasma membrane.

Prior to T cell receptor (TCR) stimulation, CD4/CD8 associated LCK remains seperated from the TCR and is maintained in an inactive state by the action of CSK. PAG bound CSK phosphorylates the negative regulatory tyrosine of LCK and inactivates the LCK kinase domain (step 1). CSK also inhibits PTPN22 by sequestering it via binding (step 2).

Upon TCR stimulation, CSK dissociates from PAG1 (step 3) and PTPN22 (step4) and is unable to inhibit LCK. Furthermore, LCK becomes activated via PTPRC-mediated dephosphorylation of negative regulatory tyrosine residues (step 5). CD4/CD8 binds MHCII receptor in APC and the associated LCK co-localizes with the TCR.

LCK is further activated by trans-autophosphorylation on the tyrosine residue on its activation loop (step 6). Active LCK further phosphorylates the tyrosine residues on CD3 chains. The signal-transducing CD3 delta/epsilon/gamma and TCR zeta chains contain a critical signaling motif known as the immunoreceptor tyrosine-based activation motif (ITAM). The two critical tyrosines of each ITAM motif are phosphorylated by LCK (step 7).

References

van Leeuwen JE & Samelson LE (1999). T cell antigen-receptor signal transduction. Curr Opin Immunol, 11, 242-8. [🔗](#)

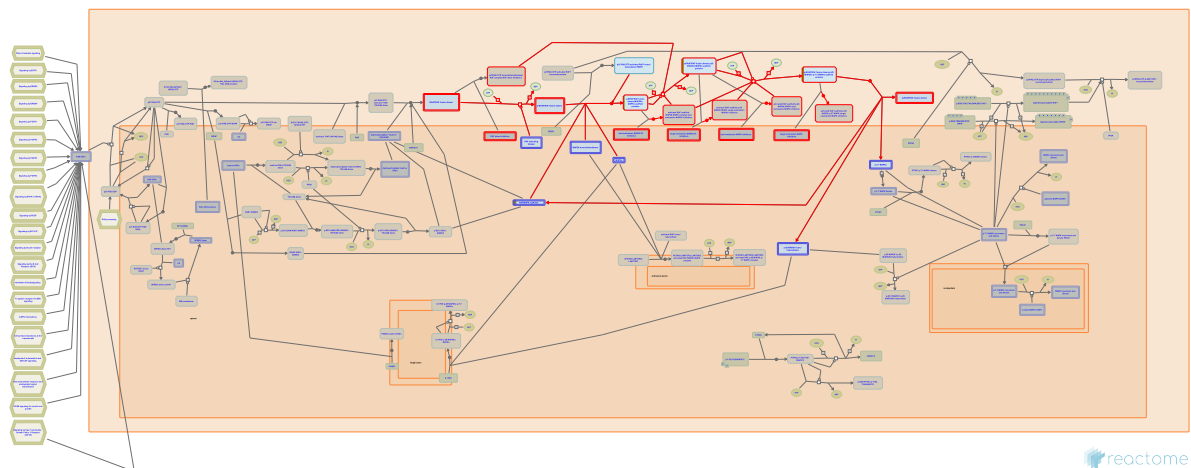
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Date	Action	Author
2007-10-29	Created	Garapati P V
2008-01-24	Authored	Rudd C.E., Garapati P V, de Bono B
2008-02-26	Reviewed	Trowsdale J
2016-05-10	Revised	Stanford S, Bottini N

3 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HLA-DQA1	P01909	HLA-DQB1	P01920	HLA-DRB5	Q30154

9. Signaling by BRAF and RAF1 fusions (R-HSA-6802952)



Diseases: cancer.

In addition to the more prevalent point mutations, BRAF and RAF1 are also subject to activation as a result of translocation events that yield truncated or fusion products (Jones et al, 2008; Cin et al, 2011; Palanisamy et al, 2010; Ciampi et al, 2005; Stransky et al, 2014; Hutchinson et al, 2013; Zhang et al, 2013; Lee et al, 2012; Ricarte-Filho et al, 2013; reviewed in Lavoie and Therrien et al, 2015). In general these events put the C-terminal kinase domain of BRAF or RAF1 downstream of an N-terminal sequence provided by a partner protein. This removes the N-terminal region of the RAF protein, relieving the autoinhibition imposed by this region of the protein. In addition, some but not all of the fusion partner proteins have been shown to contain coiled-coil or other dimerization domains. Taken together, the fusion proteins are thought to dimerize constitutively and activate downstream signaling (Jones et al, 2008; Lee et al, 2012; Hutchinson et al, 2013; Ciampi et al, 2005; Cin et al, 2011; Stransky et al, 2014).

References

Lu C, Dooling DJ, Qaddoumi I, Parker M, Ellison DW, Rusch M, ... Li Y (2013). Whole-genome sequencing identifies genetic alterations in pediatric low-grade gliomas. *Nat. Genet.*, 45, 602-12. [🔗](#)

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Kim JL, Schalm S, Stransky N, Lengauer C & Cerami E (2014). The landscape of kinase fusions in cancer. *Nat Commun*, 5, 4846. [🔗](#)

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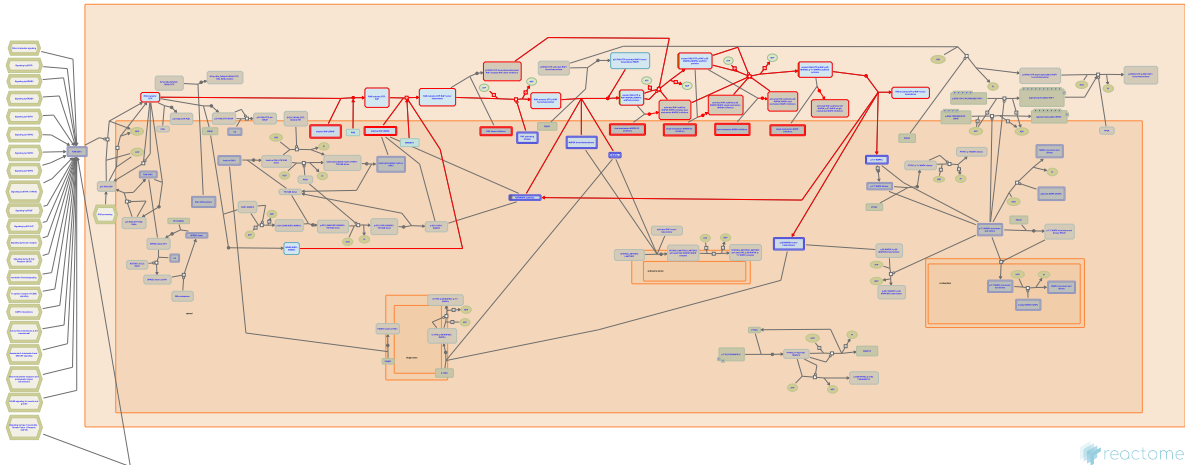
Date	Action	Author
2015-08-10	Edited	Rothfels K

Date	Action	Author
2015-08-10	Authored	Rothfels K
2015-10-02	Created	Rothfels K
2016-08-05	Reviewed	Stephens RM

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

10. Signaling downstream of RAS mutants (R-HSA-9649948)



Diseases: cancer.

Disease-causing mutations in RAS favour the active RAS:GTP bound form and yield constitutively active forms of the protein (reviewed in Prior et al, 2011; Maertens and Cichowski, 2014). Mutations in RAS contribute to cellular proliferation, transformation and survival by activating the MAPK signaling pathway, the AKT pathway and the RAL GDS pathway, among others (reviewed in Stephen et al, 2014; Pylayeva-Gupta et al, 2011)

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McCormick F, Stephen AG, Bagni RK & Esposito D (2014). Dragging ras back in the ring. Cancer Cell , 25, 272-81. [🔗](#)

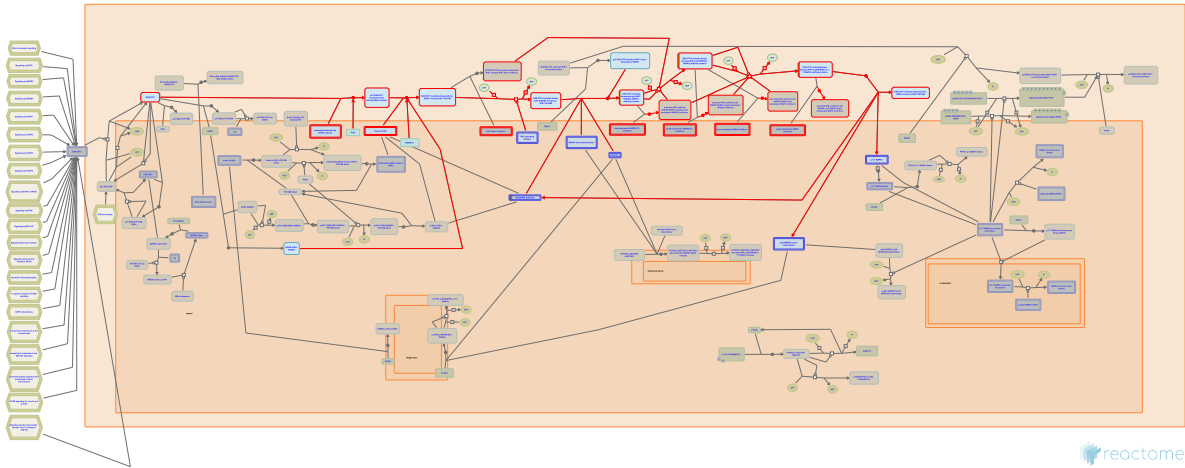
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2015-05-18	Authored	Rothfels K
2016-08-05	Reviewed	Stephens RM
2019-06-19	Created	Rothfels K

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

11. Signaling by moderate kinase activity BRAF mutants (R-HSA-6802946)



Diseases: cancer.

In addition to the highly prevalent and activating V600E BRAF mutations, numerous moderately activating and less common mutations have also been identified in human cancers (Forbes et al, 2015). Unlike the case for their highly activating counterparts, signaling through these mutant versions of BRAF depends both upon RAS binding and RAF dimerization (Wan et al, 2004; Freeman et al, 2013; Röring et al, 2012; reviewed in Lito et al, 2013; Lavoie and Therrien, 2015)

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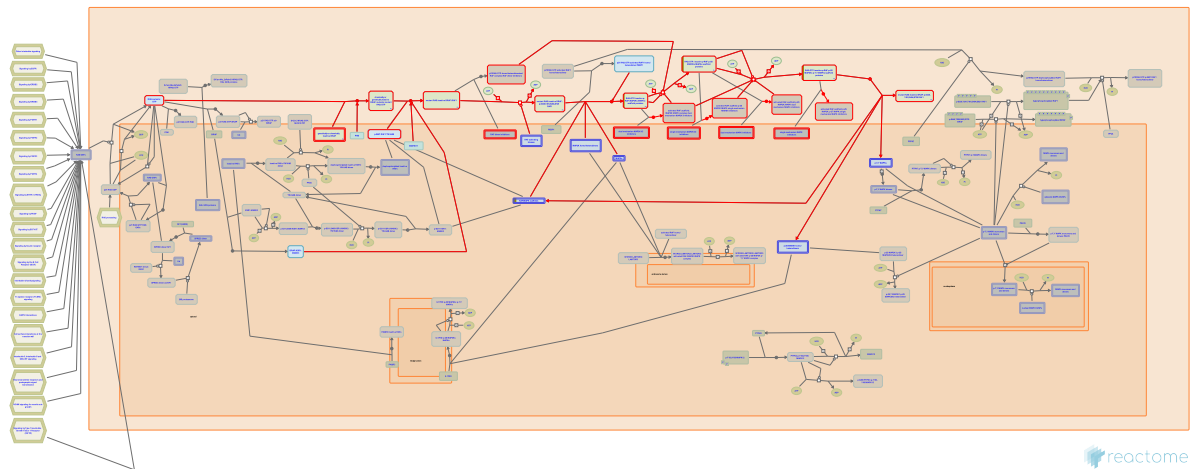
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2015-08-10	Authored	Rothfels K
2015-10-02	Created	Rothfels K
2016-08-05	Reviewed	Stephens RM

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

12. Paradoxical activation of RAF signaling by kinase inactive BRAF (R-HSA-6802955)



Diseases: cancer.

While BRAF-specific inhibitors inhibit MAPK/ERK activation in the presence of the BRAF V600E mutant, paradoxical activation of ERK signaling has been observed after treatment of cells with inhibitor in the presence of WT BRAF (Wan et al, 2004; Garnett et al, 2005; Heidorn et al, 2010; Hazivassiliou et al, 2010; Poulikakos et al, 2010). This paradoxical ERK activation is also seen in cells expressing kinase-dead or impaired versions of BRAF such as D594V, which occur with low frequency in some cancers (Wan et al, 2004; Heidorn et al, 2010). Unlike BRAF V600E, which occurs exclusively of activating RAS mutations, kinase-impaired versions of BRAF are coincident with RAS mutations in human cancers, and indeed, paradoxical activation of ERK signaling in the presence of inactive BRAF is enhanced in the presence of oncogenic RAS (Heidorn et al, 2010; reviewed in Holderfield et al, 2014). Although the details remain to be worked out, paradoxical ERK activation in the presence of inactive BRAF appears to rely on enhanced dimerization with and transactivation of CRAF (Heidorn et al, 2010; Hazivassiliou et al, 2010; Poulikakos et al, 2010; Roring et al, 2012; Rajakulendran et al, 2009; Holderfield et al, 2013; Freeman et al, 2013; reviewed in Roskoski, 2010; Samatar and Poulikakos, 2014; Lavoie and Therrien, 2015). RAF inhibitors can promote association of RAF-RAS interaction and enhanced RAF dimerization through disruption of intramolecular interactions between the kinase domain and its N-terminal regulatory region. Moreover, specific BRAF inhibitors can only occupy one protomer within the transactivated BRAF dimer due to negative co-operativity leading to paradoxical ERK activation. (Karoulia et al, 2016; Jin et al, 2017, reviewed in Karoulia et al, 2017).

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Fiala GJ, Herr R, Röring M, Heilmann K, Schamel WW, Saunders DN, ... Eisenhardt AE (2012). Distinct requirement for an intact dimer interface in wild-type, V600E and kinase-dead B-Raf signalling. EMBO J., 31, 2629-47. [🔗](#)

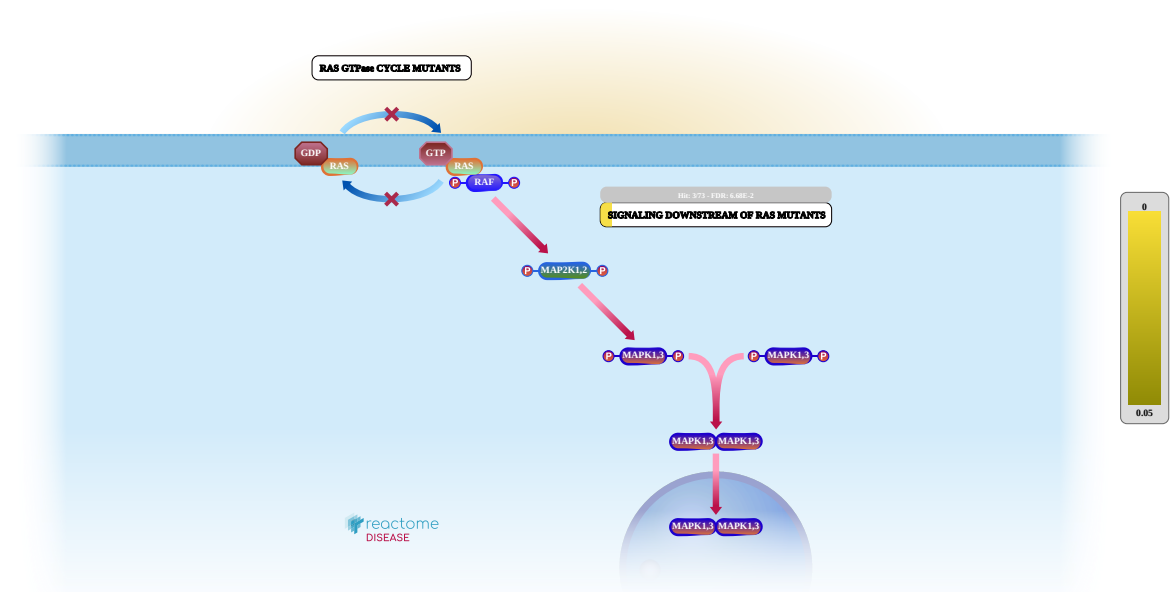
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Date	Action	Author
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2015-08-10	Authored	Rothfels K
2015-10-02	Created	Rothfels K
2016-08-05	Reviewed	Stephens RM

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

13. Signaling by RAS mutants (R-HSA-6802949)



Diseases: cancer, Noonan syndrome.

Members of the RAS gene family were the first oncogenes to be identified, and mutations in RAS are present in ~20-30% of human cancers (reviewed in Prior et al, 2012). Mutations in the KRAS gene are the most prevalent, and are found with high frequency in colorectal cancer, non-small cell lung cancer and pancreatic cancer, among others. The reasons for the lower prevalence of HRAS and NRAS mutations in human cancers are not fully understood, but may reflect gene-specific functions as well as differential codon usage and spatio-temporal regulation (reviewed in Prior et al, 2012; Stephen et al, 2014; Pylayeva-Gupta et al, 2011). Activating RAS mutations contribute to cellular proliferation, transformation and survival by activating the MAPK signaling pathway, the AKT pathway and the RAL GDS pathway, among others (reviewed in Stephen et al, 2014; Pylayeva-Gupta et al, 2011).

Although the frequency and distribution varies between RAS genes and cancer types, the vast majority of activating RAS mutations occur at one of three residues - G12, G13 and Q61. Mutations at these sites favour the RAS:GTP bound form and yield constitutively active versions of the protein (reviewed in Prior et al, 2012).

References

Grabocka E, Pylayeva-Gupta Y & Bar-Sagi D (2011). RAS oncogenes: weaving a tumorigenic web. Nat. Rev. Cancer, 11, 761-74. [🔗](#)

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McCormick F, Stephen AG, Bagni RK & Esposito D (2014). Dragging ras back in the ring. Cancer Cell, 25, 272-81. [🔗](#)

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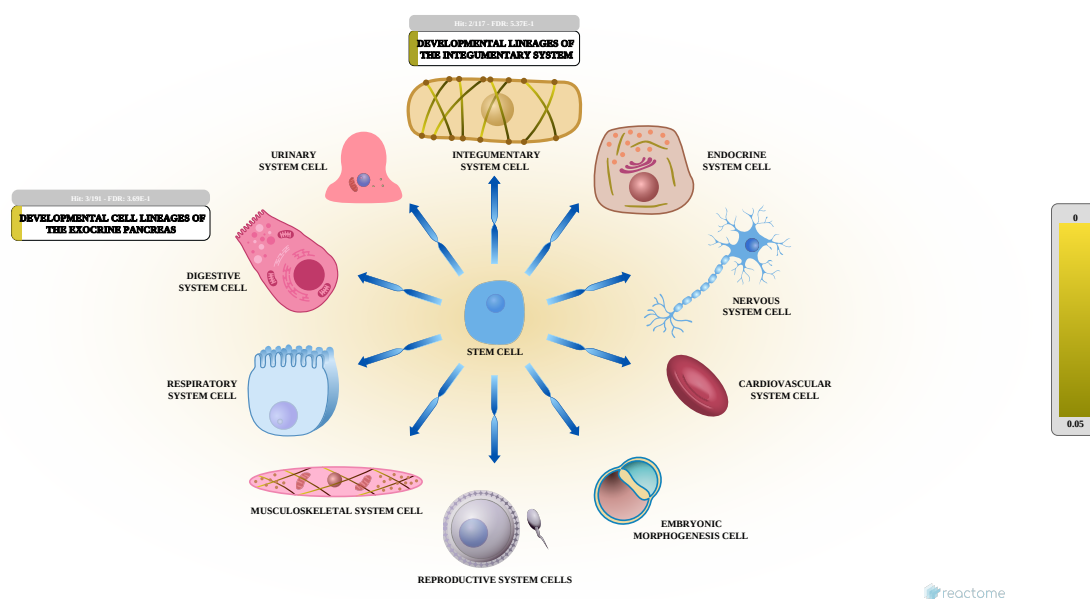
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Date	Action	Author
2015-08-10	Edited	Rothfels K
2015-10-02	Created	Rothfels K
2016-08-05	Reviewed	Stephens RM
2023-10-12	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

14. Developmental Cell Lineages (R-HSA-9734767)



Our bodies are built of >30 trillion cells specialized to fulfill diverse roles within our tissues, organs, and organ systems. All these cells originate from a single cell, a zygote formed at conception. From zygote to fetus, and throughout childhood, adolescence, and adulthood, cells divide and commit to different fates in order for the organism to develop, sustain and regenerate. The series of steps that lead from an undifferentiated progenitor cell, such as a stem cell, to one of its several possible specialized descendants constitutes a cell lineage path (Burgess et al. 2018). Cell lineage paths are organized by organ systems. Each cell lineage path cross-references a Gene Ontology (GO) biological process (The Gene Ontology Consortium 2019), and consists of a series of causally connected cell development steps. Cell development steps describe the transition between cell states during development or differentiation and are characterized by regulators (molecules promoting or inhibiting the step) and, when established, “required input components” (cell state biomarkers required for the action of regulators). Each cell state is characterized by a cell type defined in Cell Ontology (Sarntivijai et al. 2014; Osumi-Sutherland 2017), anatomical location from UBERON (Haendel et al. 2014), and a unique combination of protein and/or RNA markers with references, when available, to CellMarker (Hu et al. 2023) and PanglaoDB (Franzen et al. 2019). For a more detailed data model description, please refer to Milacic et al. 2024. Recent technological advances have allowed researchers to harvest high-throughput omics data from single cells of multicellular organisms and use it to track and manipulate cell fates (Burgess 2018; Saelens et al. 2019). This opens the door to the possibility of deciphering cell lineage paths at single-cell resolution, a critical requirement for the advancement of regenerative medicine and cancer medicine.

The cell lineage path “Differentiation of keratinocytes in interfollicular epidermis in mammalian skin” describes the differentiation of keratinocytes from stem cells to corneocytes in the interfollicular epidermis, the skin surface layer in between the adnexa (hair follicles, sweat glands, and sebaceous glands).

References

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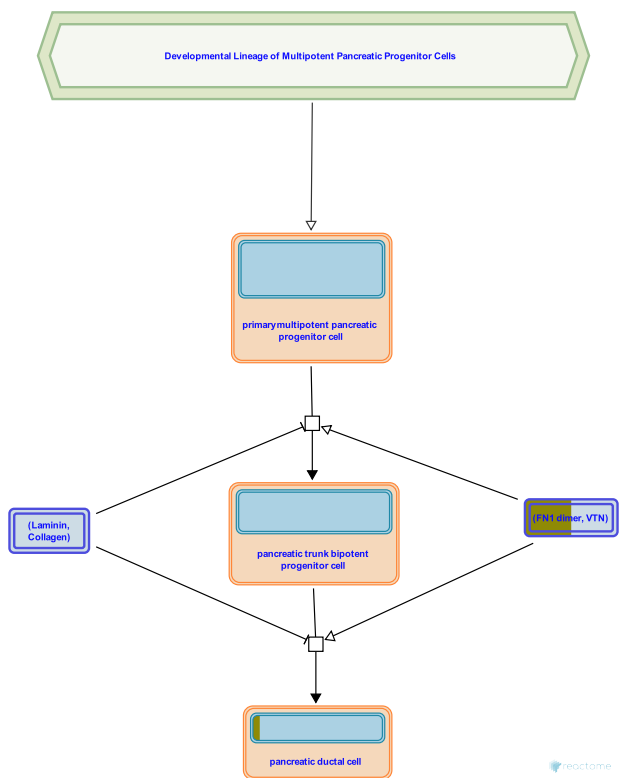
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2021-06-18	Created	Orlic-Milacic M
2024-02-28	Edited	Orlic-Milacic M
2024-02-28	Reviewed	Matthews L
2024-11-05	Reviewed	Matthews L
2025-02-26	Reviewed	Matthews L
2025-05-26	Modified	Orlic-Milacic M
2025-05-26	Reviewed	Matthews L

3 submitted entities found in this pathway, mapping to 5 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CD74	P04233	FN1	P02751	VIM	P08670

Input	Ensembl Id	Input	Ensembl Id
CD74	ENST00000009530	VIM	ENST00000544301

15. Developmental Lineage of Pancreatic Ductal Cells (R-HSA-9925563)



Ductal epithelial cells line the pancreatic ducts in a single layer, changing the shape as ducts increase in size from cuboidal, through low columnar, to high columnar epithelium (Liggitt and Dintzis, “Pancreas”, pp. 241-250). Ductal epithelial cells may have cilia or microvilli on their apical surface (Liggitt and Dintzis, “Pancreas”, pp. 241-250). Besides ductal epithelial cells, main pancreatic ducts sometimes include goblet cells (Liggitt and Dintzis, “Pancreas”, pp. 241-250).

Ductal epithelial cells develop from bipotent pancreatic trunk progenitor cells, which also give rise to endocrine progenitors and are derived from the primary multipotent pancreatic progenitor cells (reviewed in Shih et al. 2013, Jennings et al. 2015).

References

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Date	Action	Author
2024-10-17	Authored	Orlic-Milacic M
2024-10-17	Created	Orlic-Milacic M
2025-02-21	Reviewed	Li NT
2025-02-25	Edited	Orlic-Milacic M

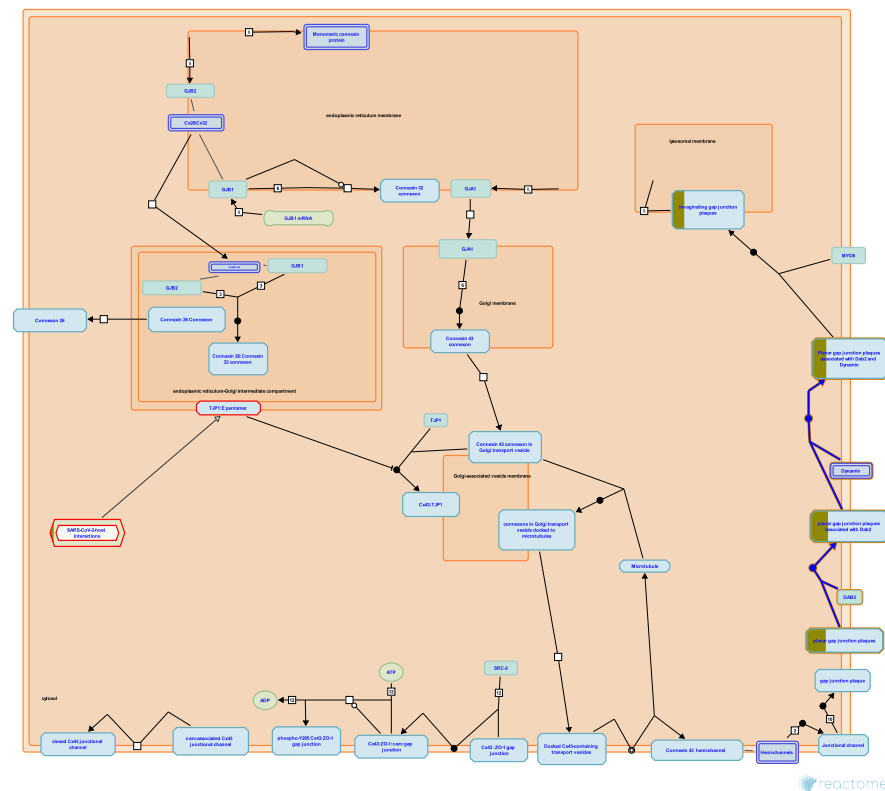
Date	Action	Author
2025-03-05	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
CD74	P04233	FN1	P02751

Input	Ensembl Id
CD74	ENST00000009530

16. Formation of annular gap junctions (R-HSA-196025)



Gap junction plaque internalization and the disruption cell communication requires a reorganization of Cx molecular interactions. Proteins including Dab-2, AP-2, Dynamin and Myosin VI associate with gap junction plaques permitting the internalisation of plaques after clathrin association (Piehl et al., 2007). Until now, two kinds of annular gap junctions have been described. The first is a small vesicle like structure which permits gap junction plaque renewal without arrest of functionality (Jordan et al., 2001). The second is a large annular structure, composed primarily of the junctional plaques of two adjacent cells (Jordan et al., 2001; Segretain et al., 2004).

References

Gumpert A, Segretain D, Falk MM, Piehl M, Denizot JP & Lehmann C (2007). Internalization of Large Double-Membrane Intercellular Vesicles by a Clathrin-dependent Endocytic Process. Mol Biol Cell. [🔗](#)

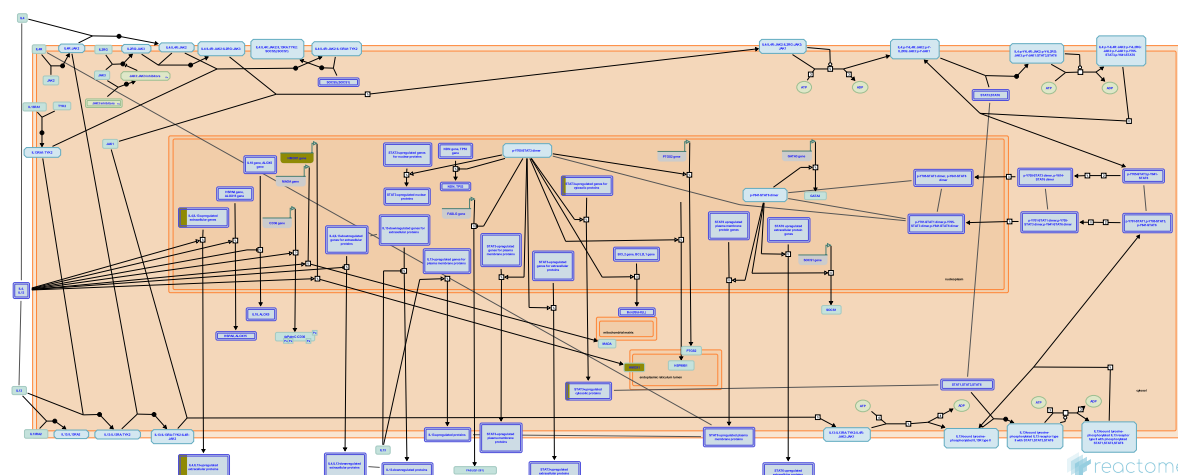
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Date	Action	Author
2007-01-03	Authored	Gilleron J, Segretain D, Falk MM
2007-04-17	Created	Matthews L
2025-06-09	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
ACTG1	P60709, P63261

17. Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)



Interleukin-4 (IL4) is a principal regulatory cytokine during the immune response, crucially important in allergy and asthma (Nelms et al. 1999). When resting T cells are antigen-activated and expand in response to Interleukin-2 (IL2), they can differentiate as Type 1 (Th1) or Type 2 (Th2) T helper cells. The outcome is influenced by IL4. Th2 cells secrete IL4, which both stimulates Th2 in an autocrine fashion and acts as a potent B cell growth factor to promote humoral immunity (Nelms et al. 1999).

Interleukin-13 (IL13) is an immunoregulatory cytokine secreted predominantly by activated Th2 cells. It is a key mediator in the pathogenesis of allergic inflammation. IL13 shares many functional properties with IL4, stemming from the fact that they share a common receptor subunit. IL13 receptors are expressed on human B cells, basophils, eosinophils, mast cells, endothelial cells, fibroblasts, monocytes, macrophages, respiratory epithelial cells, and smooth muscle cells, but unlike IL4, not T cells. Thus IL13 does not appear to be important in the initial differentiation of CD4 T cells into Th2 cells, rather it is important in the effector phase of allergic inflammation (Hershey et al. 2003).

IL4 and IL13 induce “alternative activation” of macrophages, inducing an anti-inflammatory phenotype by signaling through IL4R alpha in a STAT6 dependent manner. This signaling plays an important role in the Th2 response, mediating anti-parasitic effects and aiding wound healing (Gordon & Martinez 2010, Loke et al. 2002)

There are two types of IL4 receptor complex (Andrews et al. 2006). Type I IL4R (IL4R1) is predominantly expressed on the surface of hematopoietic cells and consists of IL4R and IL2RG, the common gamma chain. Type II IL4R (IL4R2) is predominantly expressed on the surface of nonhematopoietic cells, it consists of IL4R and IL13RA1 and is also the type II receptor for IL13. (Obiri et al. 1995, Aman et al. 1996, Hilton et al. 1996, Miloux et al. 1997, Zhang et al. 1997). The second receptor for IL13 consists of IL4R and Interleukin-13 receptor alpha 2 (IL13RA2), sometimes called Interleukin-13 binding protein (IL13BP). It has a high affinity receptor for IL13 (Kd = 250 pmol/L) but is not sufficient to render cells responsive to IL13, even in the presence of IL4R (Donaldson et al. 1998). It is reported to exist in soluble form (Zhang et al. 1997) and when overexpressed reduces JAK-STAT signaling (Kawakami et al. 2001). It's function may be to prevent IL13 signalling via the functional IL4R:IL13RA1 receptor. IL13RA2 is overexpressed and enhances cell invasion in some human cancers (Joshi & Puri 2012).

The first step in the formation of IL4R1 (IL4:IL4R:IL2RB) is the binding of IL4 with IL4R (Hoffman et al. 1995, Shen et al. 1996, Hage et al. 1999). This is also the first step in formation of IL4R2 (IL4:IL4R:IL13RA1). After the initial binding of IL4 and IL4R, IL2RB binds (LaPorte et al. 2008), to form IL4R1. Alternatively, IL13RA1 binds, forming IL4R2. In contrast, the type II IL13 complex (IL13R2) forms with IL13 first binding to IL13RA1 followed by recruitment of IL4R (Wang et al. 2009).

Crystal structures of the IL4:IL4R:IL2RG, IL4:IL4R:IL13RA1 and IL13:IL4R:IL13RA1 complexes have been determined (LaPorte et al. 2008). Consistent with these structures, in monocytes IL4R is tyrosine phosphorylated in response to both IL4 and IL13 (Roy et al. 2002, Gordon & Martinez 2010) while IL13RA1 phosphorylation is induced only by IL13 (Roy et al. 2002, LaPorte et al. 2008) and IL2RG phosphorylation is induced only by IL4 (Roy et al. 2002).

Both IL4 receptor complexes signal through Jak/STAT cascades. IL4R is constitutively-associated with JAK2 (Roy et al. 2002) and associates with JAK1 following binding of IL4 (Yin et al. 1994) or IL13 (Roy et al. 2002). IL2RG constitutively associates with JAK3 (Boussiotis et al. 1994, Russell et al. 1994). IL13RA1 constitutively associates with TYK2 (Umeshita-Suyama et al. 2000, Roy et al. 2002, LaPorte et al. 2008, Bhattacharjee et al. 2013).

IL4 binding to IL4R1 leads to phosphorylation of JAK1 (but not JAK2) and STAT6 activation (Takeda et al. 1994, Ratthe et al. 2007, Bhattacharjee et al. 2013).

IL13 binding increases activating tyrosine-99 phosphorylation of IL13RA1 but not that of IL2RG. IL4 binding to IL2RG leads to its tyrosine phosphorylation (Roy et al. 2002). IL13 binding to IL4R2 leads to TYK2 and JAK2 (but not JAK1) phosphorylation (Roy & Cathcart 1998, Roy et al. 2002).

Phosphorylated TYK2 binds and phosphorylates STAT6 and possibly STAT1 (Bhattacharjee et al. 2013).

A second mechanism of signal transduction activated by IL4 and IL13 leads to the insulin receptor substrate (IRS) family (Kelly-Welch et al. 2003). IL4R1 associates with insulin receptor substrate 2 and activates the PI3K/Akt and Ras/MEK/Erk pathways involved in cell proliferation, survival and translational control. IL4R2 does not associate with insulin receptor substrate 2 and consequently the PI3K/Akt and Ras/MEK/Erk pathways are not activated (Busch-Dienstfertig & González-Rodríguez 2013).

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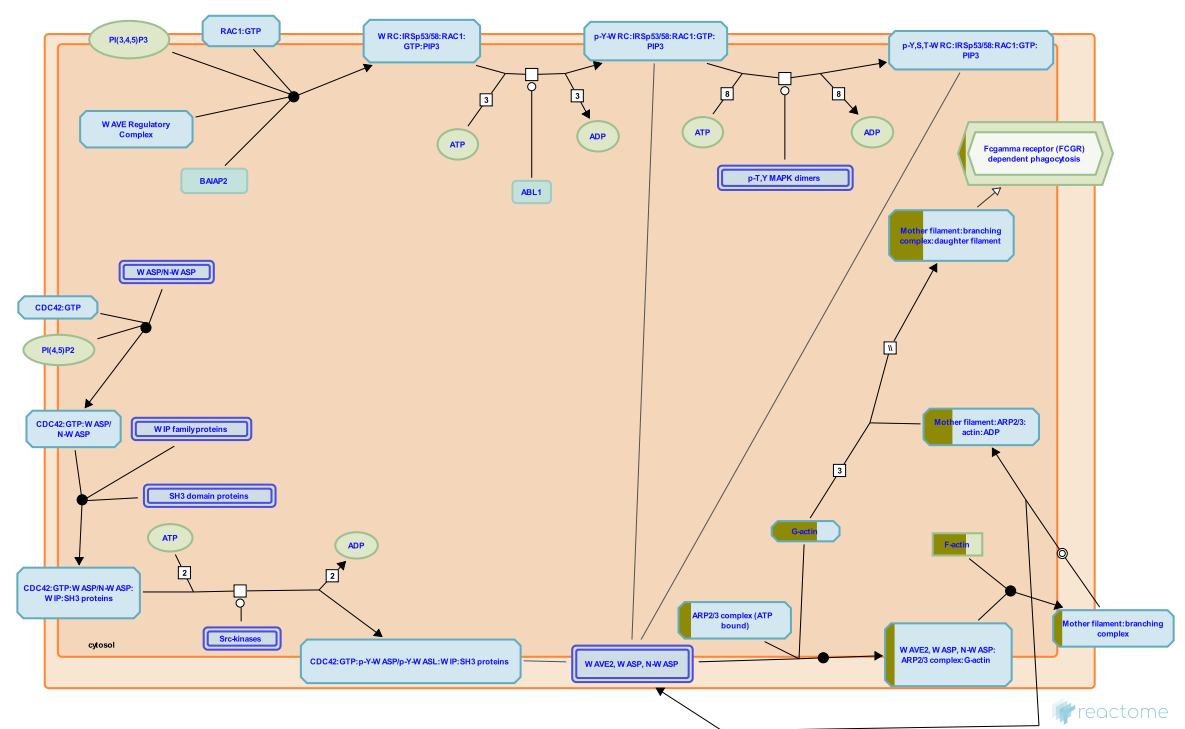
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2015-07-01	Created	Jupe S
2016-09-02	Edited	Jupe S
2016-09-02	Reviewed	Leibovich SJ

3 submitted entities found in this pathway, mapping to 6 Reactome entities

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Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
FN1	ENSG00000115414	HMOX1	ENSG00000100292	VIM	ENSG00000026025

18. RHO GTPases Activate WASPs and WAVES (R-HSA-5663213)



WASP and WAVE proteins belong to the Wiskott-Aldrich Syndrome protein family, with recessive mutations in the founding member WASP being responsible for the X-linked recessive immunodeficiency known as the Wiskott-Aldrich Syndrome. WASP proteins include WASP and WASL (N-WASP). WAVE proteins include WASF1 (WAVE1), WASF2 (WAVE2) and WASF3 (WAVE3). WASPs and WAVES contain a VCA domain (consisting of WH2 and CA subdomains) at the C-terminus, responsible for binding to G-actin (WH2 subdomain) and the actin-associated ARP2/3 complex (CA subdomain). WASPs contain a WH1 (WASP homology 1) domain at the N-terminus, responsible for binding to WIPs (WASP-interacting proteins). A RHO GTPase binding domain (GBD) is located in the N-terminal half of WASPs and C-terminally located in WAVES. RHO GTPases activate WASPs by disrupting the autoinhibitory interaction between the GBD and VCA domains, which allows WASPs to bind actin and the ARP2/3 complex and act as nucleation promoting factors in actin polymerization. WAVES have the WAVE/SCAR homology domain (WHD/SHD) at the N-terminus, which binds ABI, NCKAP1, CYFIP2 and BRK1 to form the WAVE regulatory complex (WRC). Binding of the RAC1:GTP to the GBD of WAVES most likely induces a conformational change in the WRC that allows activating phosphorylation of WAVES by ABL1, thus enabling them to function as nucleation promoting factors in actin polymerization through binding G-actin and the ARP2/3 complex (Reviewed by Lane et al. 2014).

References

Jiang WG, Martin T, Weeks HP & Lane J (2014). Structure and role of WASP and WAVE in Rho GTPase signalling in cancer. *Cancer Genomics Proteomics*, 11, 155-65. [🔗](#)

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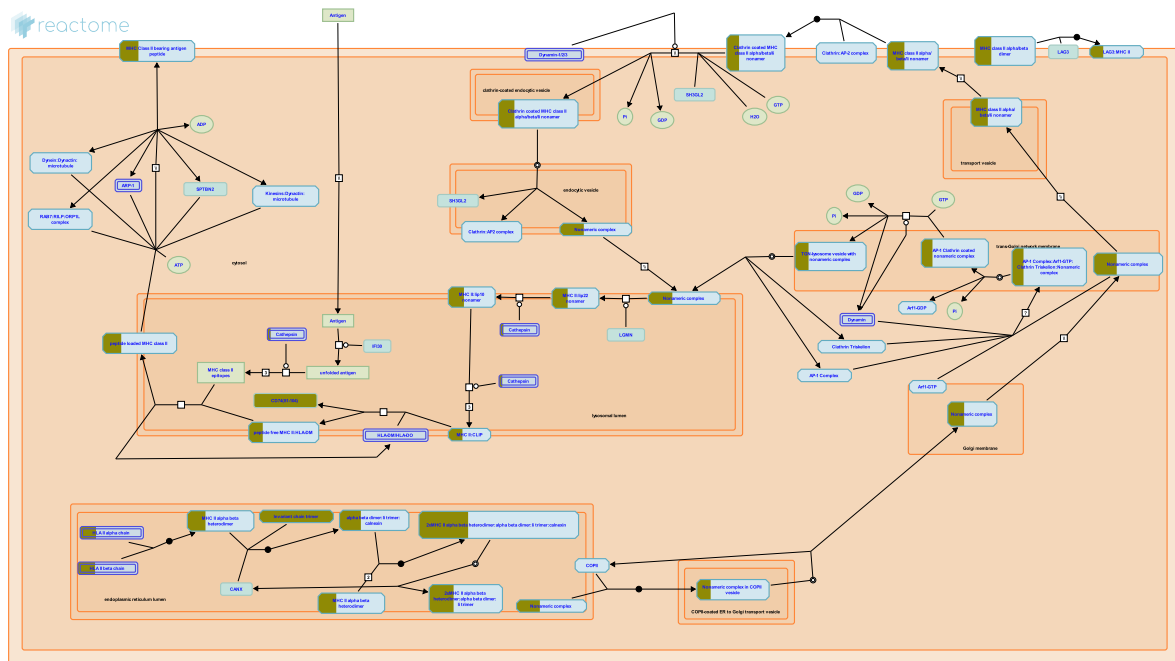
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2014-12-26	Authored	Rivero Crespo F

Date	Action	Author
2015-01-16	Created	Orlic-Milacic M
2015-02-02	Edited	Orlic-Milacic M

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	ARPC1B	O15143

19. MHC class II antigen presentation (R-HSA-2132295)



Antigen presenting cells (APCs) such as B cells, dendritic cells (DCs) and monocytes/macrophages express major histocompatibility complex class II molecules (MHC II) at their surface and present exogenous antigenic peptides to CD4⁺ T helper cells. CD4⁺ T cells play a central role in immune protection. On their activation they stimulate differentiation of B cells into antibody-producing B-cell blasts and initiate adaptive immune responses. MHC class II molecules are transmembrane glycoprotein heterodimers of alpha and beta subunits. Newly synthesized MHC II molecules present in the endoplasmic reticulum bind to a chaperone protein called invariant (Ii) chain. The binding of Ii prevents the premature binding of self antigens to the nascent MHC molecules in the ER and also guides MHC molecules to endocytic compartments. In the acidic endosomal environment, Ii is degraded in a stepwise manner, ultimately to free the class II peptide-binding groove for loading of antigenic peptides. Exogenous antigens are internalized by the APC by receptor mediated endocytosis, phagocytosis or pinocytosis into endocytic compartments of MHC class II positive cells, where engulfed antigens are degraded in a low pH environment by multiple acidic proteases, generating MHC class II epitopes. Antigenic peptides are then loaded into the class II ligand-binding groove. The resulting class II peptide complexes then move to the cell surface, where they are scanned by CD4⁺ T cells for specific recognition (Berger & Roche 2009, Zhou & Blum 2004, Watts 2004, Landsverk et al. 2009).

References

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Simmons D, Harding CV & Ramachandra L (2009). MHC molecules and microbial antigen processing in phagosomes. Curr Opin Immunol, 21, 98-104. [🔗](#)

Edit history

Date	Action	Author
2012-02-21	Edited	Garapati P V
2012-02-21	Authored	Garapati P V
2012-02-21	Created	Garapati P V
2012-05-14	Reviewed	Neeffes J
2025-06-09	Modified	Weiser JD

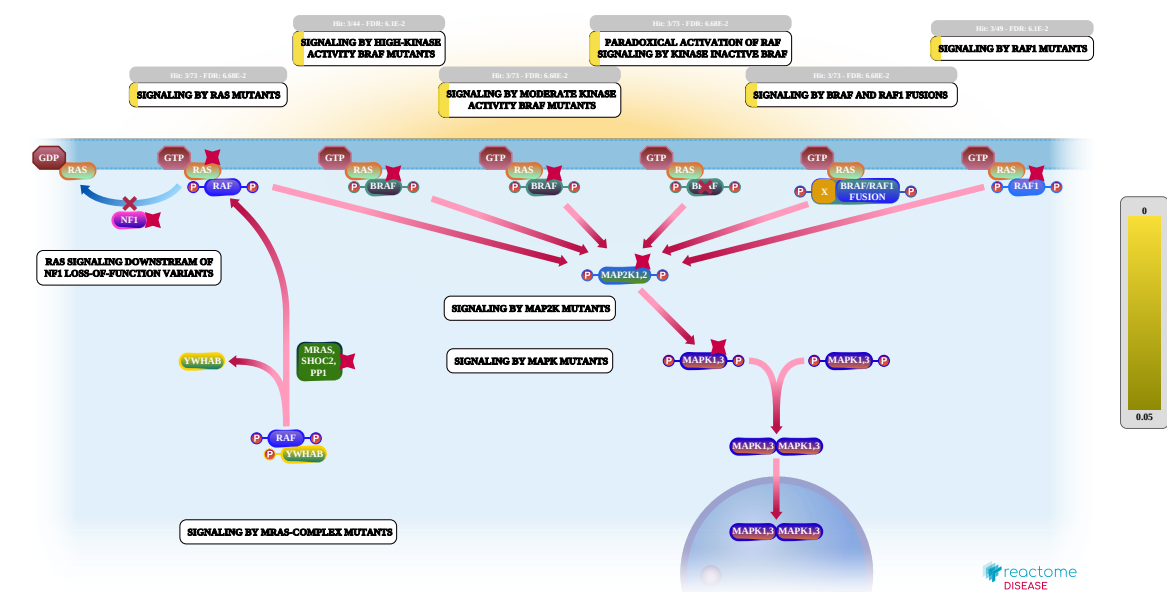
5 submitted entities found in this pathway, mapping to 5 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CD74	P04233	CTSD	P07339	HLA-DQA1	P01909
HLA-DQB1	P01920	HLA-DRB5	Q30154		

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HLA-DQA1	P01909	P04233			

20. Oncogenic MAPK signaling ([R-HSA-6802957](#))



Diseases: cancer.

The importance of the RAS/RAF/MAPK cascade in regulating cellular proliferation, differentiation and survival is highlighted by the fact that components of the pathway are mutated with high frequency in a large number of human cancers. Activating mutations in RAS are found in approximately one third of human cancers, while ~8% of tumors express an activated form of BRAF. RAS pathway activation is also achieved in a smaller subset of cancers by loss-of-function mutations in negative regulators of RAS signaling, such as the RAS GAP NF1 (reviewed in Prior et al, 2012; Pylayeva-Gupta et al, 2011; Stephen et al, 2014; Lavoie and Therrien, 2015; Lito et al, 2013; Samatar and Poulikakos, 2014; Maertens and Cichowski, 2014).

References

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Grabocka E, Pylayeva-Gupta Y & Bar-Sagi D (2011). RAS oncogenes: weaving a tumorigenic web. *Nat. Rev. Cancer*, 11, 761-74. [🔗](#)

Cichowski K & Maertens O (2014). An expanding role for RAS GTPase activating proteins (RAS GAPs) in cancer. *Adv Biol Regul*, 55, 1-14. [🔗](#)

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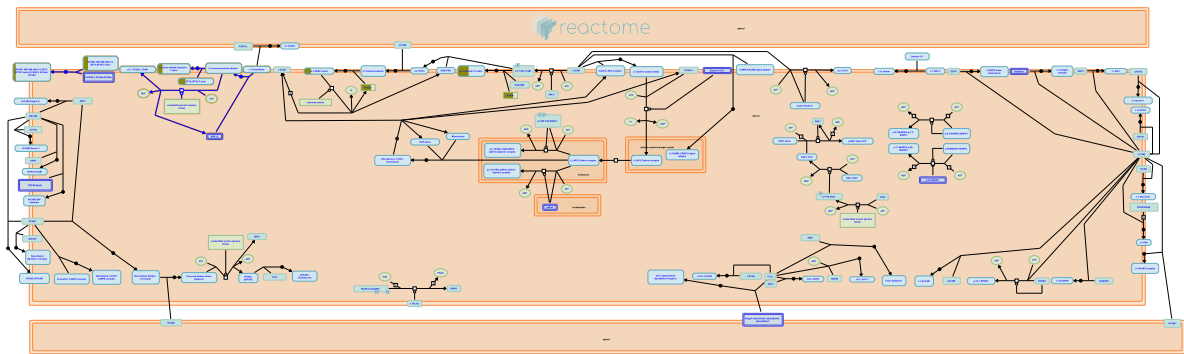
Date	Action	Author
2015-05-18	Edited	Rothfels K
2015-05-18	Authored	Rothfels K

Date	Action	Author
2015-10-02	Created	Rothfels K
2016-08-05	Reviewed	Stephens RM
2020-05-04	Reviewed	Gavathiotis E
2023-10-12	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	FN1	P02751

21. Interaction between L1 and Ankyrins (R-HSA-445095)



Ankyrins are a family of adaptor proteins that couple membrane proteins such as voltage gated Na⁺ channels and the Na⁺/K⁺ anion exchanger to the spectrin actin cytoskeleton. Ankyrins are encoded by three genes (ankyrin-G, -B and -R) of which ankyrin-G and -B are the major forms expressed in the developing nervous system. Ankyrins bind to the cytoplasmic domain of L1 CAMs and couple them and ion channel proteins, to the spectrin cytoskeleton. This binding enhances the homophilic adhesive activity of L1 and reduces its mobility within the plasma membrane. L1 interaction with ankyrin mediates branching and synaptogenesis of cortical inhibitory neurons.

References

Nagaraj K, Hortsch M & Godenschwege TA (2009). The interaction between L1-type proteins and ankyrins--a master switch for L1-type CAM function. *Cell Mol Biol Lett*, 14, 57-69. [🔗](#)

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Maness PF & Guan H (2010). Perisomatic GABAergic Innervation in Prefrontal Cortex Is Regulated by Ankyrin Interaction with the L1 Cell Adhesion Molecule. *Cereb Cortex*. [🔗](#)

Gunn-Moore FJ, Hill M, Herron LR & Davey F (2009). The intracellular interactions of the L1 family of cell adhesion molecules. *Biochem J*, 419, 519-31. [🔗](#)

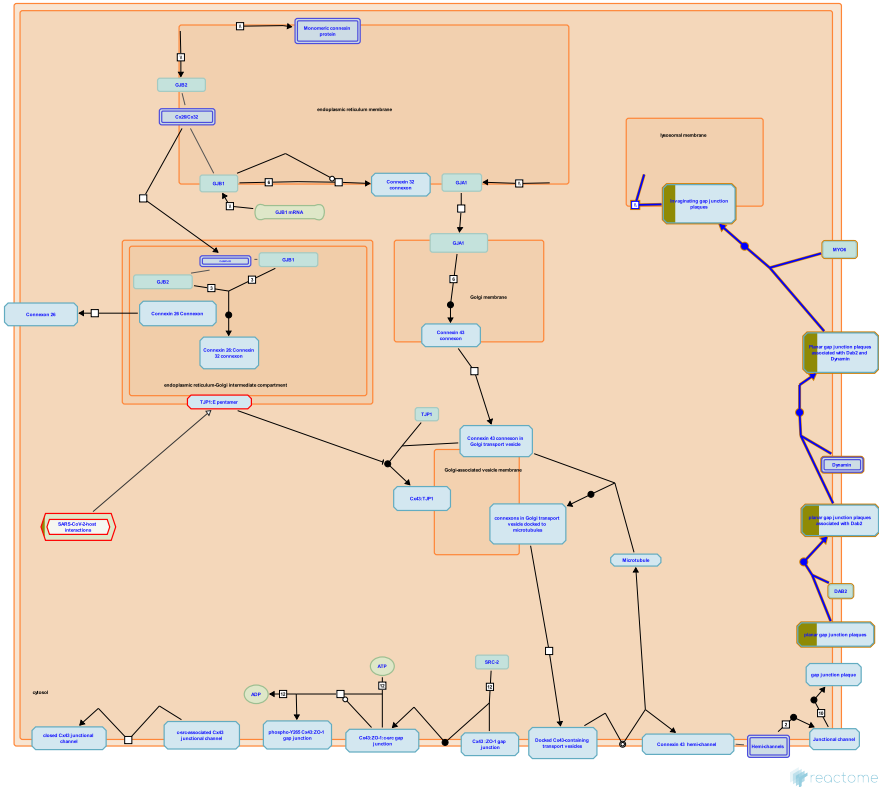
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Date	Action	Author
2008-07-30	Edited	Garapati P V
2008-07-30	Authored	Garapati P V
2009-10-27	Created	Garapati P V
2010-02-16	Reviewed	Maness PF
2025-06-08	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
ACTG1	P60709, P63261

22. Gap junction degradation (R-HSA-190873)



The half-life of Cx is very short (1 to 5h) compared to other junctional proteins (Laird et al., 1995 ; Fallon and Goudenough, 1981). Connexins are targeted for degradation by the proteasome and the lysosome. Degradation appears to involve the phosphorylation of Connexins as well as their interactions with other proteins (Piehl et al., 2007).

References

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Kasprzak L, Laird DW & Castillo M (1995). Gap junction turnover, intracellular trafficking, and phosphorylation of connexin43 in brefeldin A-treated rat mammary tumor cells. J Cell Biol, 131, 1193-203. [🔗](#)

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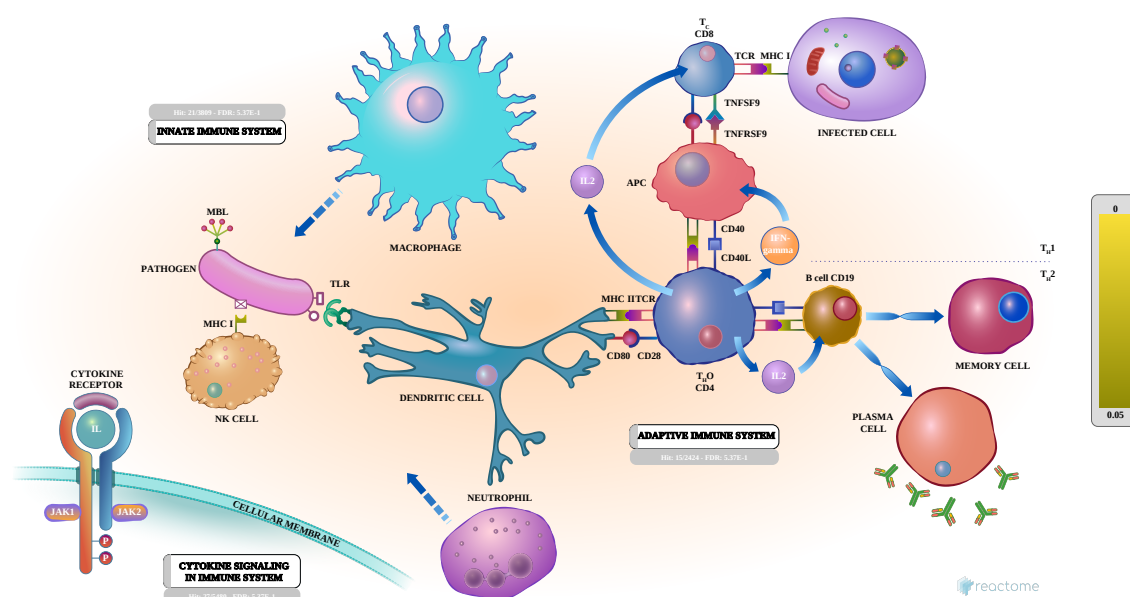
Date	Action	Author
2007-01-03	Authored	Gilleron J, Segretain D, Falk MM
2007-01-09	Created	Matthews L
2007-03-27	Edited	Matthews L
2025-06-09	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
ACTG1	P60709, P63261

Input	UniProt Id
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23. Immune System (R-HSA-168256)



Humans are exposed to millions of potential pathogens daily, through contact, ingestion, and inhalation. Our ability to avoid infection depends on the adaptive immune system and during the first critical hours and days of exposure to a new pathogen, our innate immune system.

References

Edit history

Date	Action	Author
2005-11-12	Created	Gillespie ME
2006-03-30	Authored	Luo F, Ouwehand WH, Gillespie ME, de Bono B
2006-04-19	Reviewed	Zwaginga JJ, D'Eustachio P, Gay NJ, Gale M Jr
2025-06-09	Modified	Weiser JD

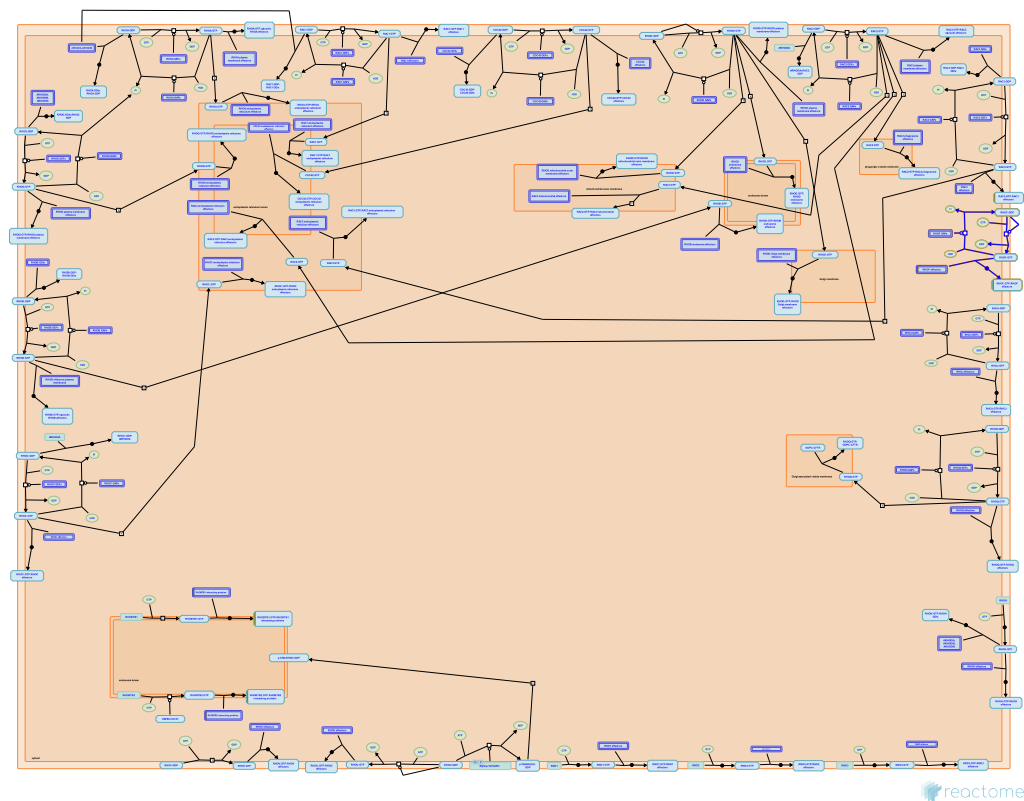
21 submitted entities found in this pathway, mapping to 32 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	ARPC1B	O15143	C1QC	P02747
CCL5	P13501	CD68	P34810	CD74	P04233
CD99	P14209	CFD	P00746	CTSD	P07339
FN1	P02751	FTH1	P02794	HLA-A	P04439
HLA-B	P01889	HLA-DQA1	P01909	HLA-DQB1	P01920
HLA-DRB5	Q30154	HMOX1	P09601	IL32	P24001
PCBP2	Q15366	PLD3	Q8IV08, Q96BZ4	VIM	P08670
Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CCL5	ENSG00000271503	FN1	ENSG00000115414	HLA-A	ENSG00000206503
HLA-B	ENSG00000234745	HLA-DQA1	ENSG00000196735	HLA-DQB1	ENSG00000179344
HLA-DRB5	ENSG00000198502	HMOX1	ENSG00000100292	VIM	ENSG00000026025

Interactors found in this pathway (15)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ACTG1	P63261	P23528, Q9Y6K9, P10636-8	APOE	P02649	P26441, Q8TBC4, P08603, Q9NZC2, P02748, Q9BQ95, Q9UBD9, Q53EL6, P37840, P02795
ARPC1B	O15143	P42768	CCL5	EBI-10021813	Q04206
CD74	P04233	Q14974	FN1	P02751-7, P02751	Q6ISS4, P06241
FTH1	P02794	O60437	GAPDH	P04406	P17612, P06241, P10599, Q9UHD2, P35222, P10636-8, P00441
HLA-A	Q5SUL5, P04439	O15533, P61769, P07225	HLA-B	P01889, I4ENP2	P43629, P61769
HLA-DQA1	P01909	P04233	HMOX1	P09601	Q16623, Q9P296
IL32	P24001-2, P24001-4, C6GKH1	Q05655, P24158	PLD3	Q8IV08	P19838
VIM	P84198, P08670	O60437, Q9HC29, P63104			

24. RHOF GTPase cycle (R-HSA-9035034)



This pathway catalogues RHOF (RIF) GTPase activator proteins (GAPs) and RHOF effectors. RHOF GTPase is thought to exist in the active GTP-bound state in the absence of any GEF activity (Tian et al. 2017). No GDP dissociation inhibitors (GDIs) have been shown to interact with RHOF. RHOF is only found in vertebrates (Gad and Aspenstrom 2010; Fan and Mellor 2012). RHOF regulates cytoskeletal dynamics (Gad and Aspenstrom 2010; Aspenstrom et al. 2014) and promotes the formation of filopodia and stress fibers (Fan and Mellor 2012). RHOF may be involved in actin remodelling in lymphocyte microvilli (Fan and Mellor 2012). In neurons, RHOF contributes to the formation of dendritic spines (Fan and Mellor 2012).

References

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Gad AK & Aspenström P (2010). Rif proteins take to the RhoD: Rho GTPases at the crossroads of actin dynamics and membrane trafficking. *Cell. Signal.*, 22, 183-9. [🔗](#)

Wang J, Tian X, Yan H, Wu S, Fan L & Li J (2017). Neurotrophin Promotes Neurite Outgrowth by Inhibiting Rif GTPase Activation Downstream of MAPKs and PI3K Signaling. *Int J Mol Sci*, 18. [🔗](#)

Mellor H & Fan L (2012). The small Rho GTPase Rif and actin cytoskeletal remodelling. *Biochem. Soc. Trans.*, 40, 268-72. [🔗](#)

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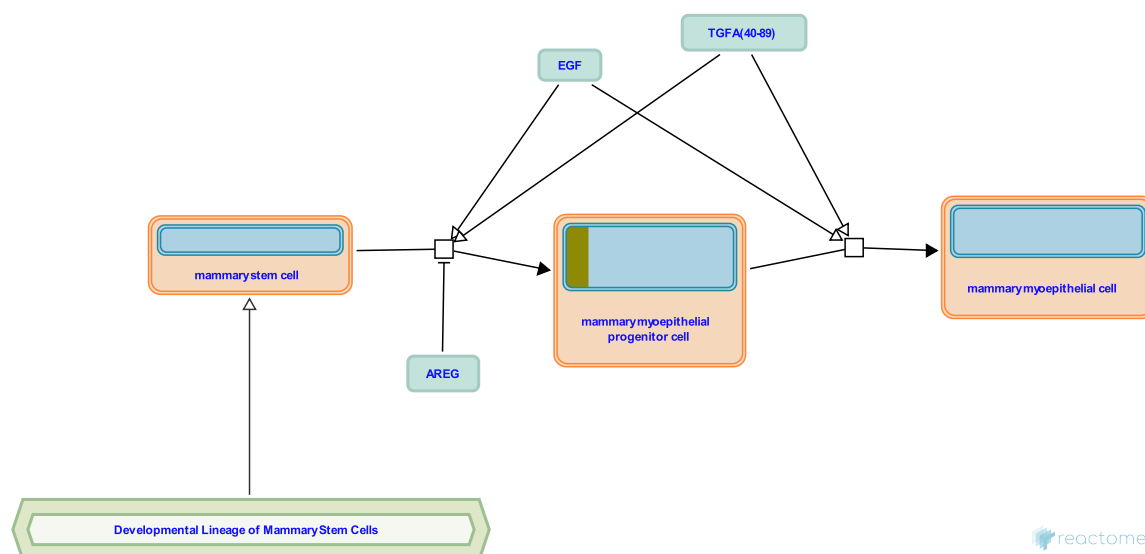
Date	Action	Author
2018-01-16	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M

Date	Action	Author
2021-02-05	Reviewed	Fort P
2021-02-20	Reviewed	Bisbal M
2021-02-25	Edited	Orlic-Milacic M
2021-03-26	Revised	Orlic-Milacic M
2021-03-26	Edited	Orlic-Milacic M
2025-06-09	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
ACTG1	P60709, Q6S8J3

25. Developmental Lineage of Mammary Gland Myoepithelial Cells (R-HSA-9927432)



Most postnatal mammary gland development originates from unipotent lineage-committed progenitors (luminal progenitor cells and myoepithelial progenitor cells), which are located in the basal epithelium (reviewed in Inman et al. 2015, Edwards and Brennan 2021).

Myoepithelial cells, which originate from myoepithelial progenitors, are located between luminal cells that line the mammary ducts and alveoli, and the basement membrane (reviewed in Stingl et al. 2005). Myoepithelial cells form a continuous sheath around the ducts but are sparser around the alveoli, where their cytoplasmic processes create a basket-like structure that allows some luminal cells to come in contact with the basement membrane (reviewed in Stingl et al. 2005). Myoepithelial cells are spindle shaped and are able to contract in response to oxytocin, which is necessary for milk secretion (lactation) (reviewed in Stingl et al. 2005, and Watson and Khaled 2020).

Upon ablation of the luminal epithelial cells in the adult mouse mammary gland, unipotent myoepithelial progenitor cells can convert into unipotent luminal progenitors to repopulate the luminal lineage (Centonze et al. 2020, reviewed in Edwards and Brennan 2021). In mouse mammary glands, tumor necrosis factor (TNF), secreted by luminal cells, restricts multipotency of myoepithelial progenitors under normal physiological conditions (Centonze et al. 2020). After ablation of luminal cells, Notch, Wnt and EGFR signaling pathways are activated in myoepithelial progenitor cells, promoting regeneration-induced multipotency of myoepithelial progenitors (Centonze et al. 2020, reviewed in Edwards and Brennan 2021). In human mammary organoids, high intensity EGFR signaling promotes myoepithelial cell fate (Pasic et al. 2011, Mukhopadhyay et al. 2013). FGF2 and FGF7 are not necessary for the differentiation of myoepithelial cells in human mammary organoids, but they are needed for the establishment of proper architecture of mammary ducts (Pasic et al. 2011).

References

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- Watson CJ & Khaled WT (2020). Mammary development in the embryo and adult: new insights into the journey of morphogenesis and commitment. *Development*, 147. [🔗](#)

Macara IG, Moskaluk CA, Brenin DR, Eisinger-Mathason TS, Lannigan DA, Velayudhan BT & Pasic L (2011). Sustained activation of the HER1-ERK1/2-RSK signaling pathway controls myoepithelial cell fate in human mammary tissue. *Genes Dev*, 25, 1641-53. [🔗](#)

Mott JD, Robertson C, Bissell MJ & Inman JL (2015). Mammary gland development: cell fate specification, stem cells and the microenvironment. *Development*, 142, 1028-42. [🔗](#)

Brennan K & Edwards A (2021). Notch Signalling in Breast Development and Cancer. *Front Cell Dev Biol*, 9, 692173. [🔗](#)

Edit history

Date	Action	Author
2024-10-15	Authored	Qiao J
2024-11-05	Created	Orlic-Milacic M
2025-05-21	Reviewed	Qiao J
2025-05-28	Edited	Orlic-Milacic M

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
VIM	P08670

Input	Ensembl Id
VIM	ENST00000544301

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

26 of the submitted entities were found, mapping to 45 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACTG1	P60709, P63261	APOE	P02649	ARPC1B	O15143
C1QC	P02747	CCL5	P13501	CD68	P34810
CD74	P04233	CD99	P14209	CFD	P00746
CTSD	P07339	FN1	P02751	FTH1	P02794
GAPDH	P04406	HLA-A	P04439	HLA-B	P01889
HLA-DQA1	P01909	HLA-DQB1	P01920	HLA-DRB5	Q30154
HMOX1	P09601	IL32	P24001	PCBP2	Q15366
PLD3	Q8IV08, Q96BZ4	RPL41	P62945	TAGLN2	P37802
UCP2	P55851	VIM	P08670		

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
APOE	ENSG00000130203	CCL5	ENSG00000271503	CD74	ENST00000009530
CTSD	ENSG00000117984	FN1	ENSG00000115414	FTH1	ENST00000273550
HLA-A	ENSG00000206503	HLA-B	ENSG00000234745	HLA-DQA1	ENSG00000196735
HLA-DQB1	ENSG00000179344	HLA-DRB5	ENSG00000198502	HMOX1	ENSG00000100292
VIM	ENST00000544301				

Interactors (24)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ACTG1	P63261	P23528, Q9Y6K9, P10636-8	APOE	P02649	P02795
ARPC1B	O15143	P42768	CCL5	EBI-10021813	Q04206
CD68	P34810	Q15058	CD74	P04233	Q14974
CRIP1	P50238	P04183	CTSD	P07339	Q8NBJ7
FN1	P02751-7, P02751	Q6ISS4, P06241	FTH1	P02794	O60437
GAPDH	P04406	P17612, P06241, P10599, Q9UHD2, P35222, P10636-8, P00441	HLA-A	P04439	P61769
HLA-B	P01889	P61769	HLA-DQA1	P01909	P04233
HLA-DRB5	Q30154	O14757	HMOX1	P09601	Q16623, Q9P296
IL32	P24001-2, P24001-4, C6GKH1	Q05655, P24158	PCBP2	Q15366	P11940
PLD3	Q8IV08	P19838	RPL41	P62945	P63244
SH3BGRL3	Q9H299	P00533	TAGLN2	P37802	P42858
UCP2	P55851	P20930	VIM	P84198, P08670	O60437, Q9HC29, P63104

7. Identifiers not found

These 2 identifiers were not found neither mapped to any entity in Reactome.

CCL18 MTRNR2L12